Programme for today

- Introductory lecture — wrangling with a question in mind
- Guided coding: *dplyr* on the COVID data
- Task 1 — build the analysis table
- Lecture — making the comparison visible
- Guided coding: *ggplot2*
- Task 2 — one analysis figure
- Peer review on the Padlet board

09:00 – 11:30

Lecture, guided coding, task 1

11:30 – 13:00

Lunch

13:00 – 15:30

Lecture, guided coding, task 2, review

At the end of the day, you should be able to take a raw table to an analysis table and a figure

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code,continent,location,date,total_cases,new_cases,new_cases_smoothed,total_deaths,new_deaths,new_deaths_smoothed,total_cases_per_million,new_cases_per_million,new_cases_smoothed_per_million,total_deaths_per_million,new_deaths_per_million,new_deaths_smoothed_per_mil
roduction_rate,icu_patients,icu_patients_per_million,hosp_patients,hosp_patients_per_million,weekly_icu_admissions,weekly_icu_admissions_per_million,weekly_hosp_admissions,weekly_hosp_admissions_per_million,new_tests,total_tests,total_tests_per_thousand,new_tests_pe
nd,new_tests_smoothed,new_tests_smoothed_per_thousand,positive_rate,tests_per_case,tests_units,total_vaccinations,people_vaccinated,people_fully_vaccinated,total_boosters,new_vaccinations,new_vaccinations_smoothed,total_vaccinations_per_hundred,people_vaccinated_per
,people_fully_vaccinated_per_hundred,total_boosters_per_hundred,new_vaccinations_smoothed_per_million,new_people_vaccinated_smoothed,new_people_vaccinated_smoothed_per_hundred,stringency_index,population,population_density,median_age,aged_65_older,aged_70_older,gdp_
ta,extreme_poverty,cardiovasc_death_rate,diabetes_prevalence,female_smokers,male_smokers,handwashing_facilities,hospital_beds_per_thousand,life_expectancy,human_development_index,excess_mortality_cumulative_absolute,excess_mortality_cumulative,excess_mortality,exces
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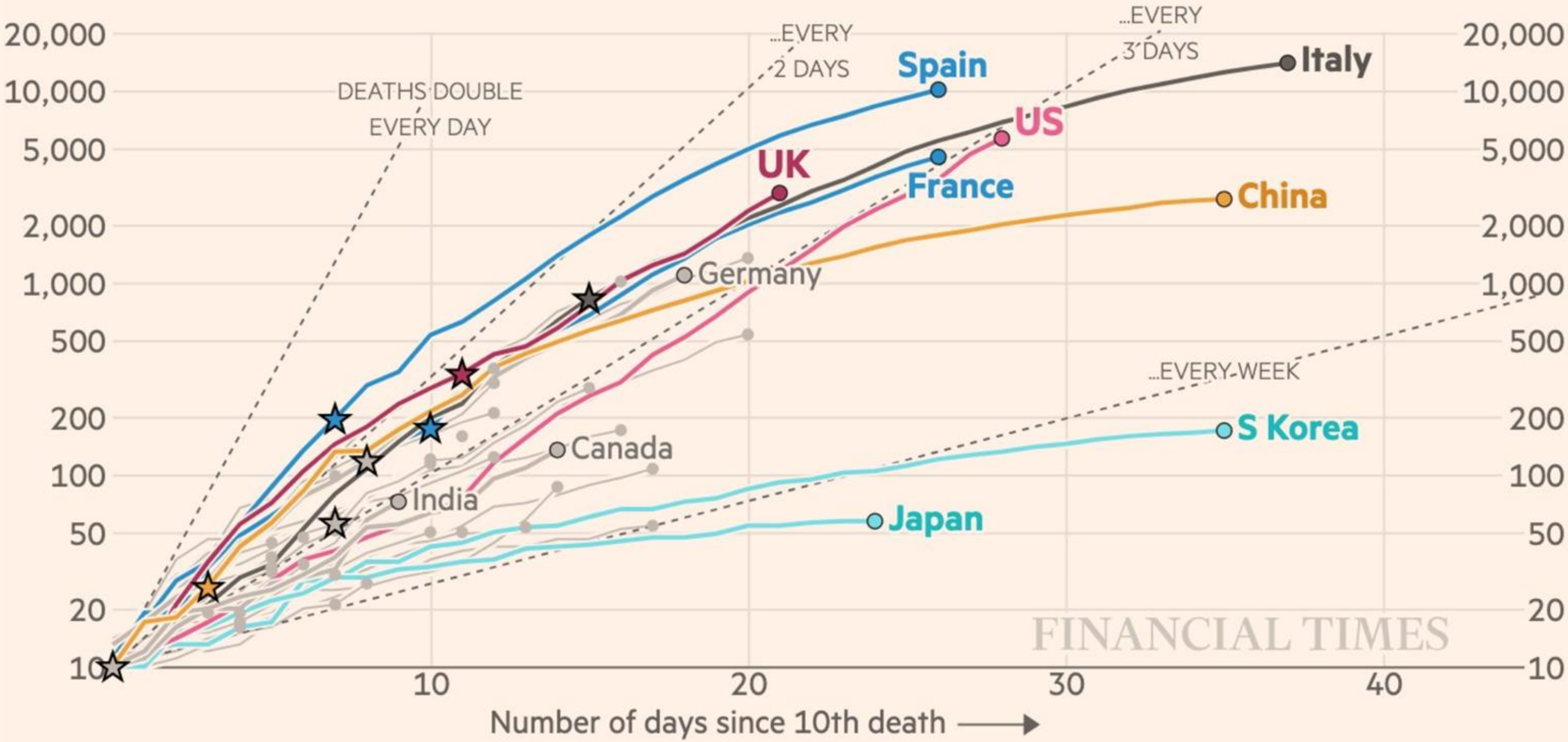
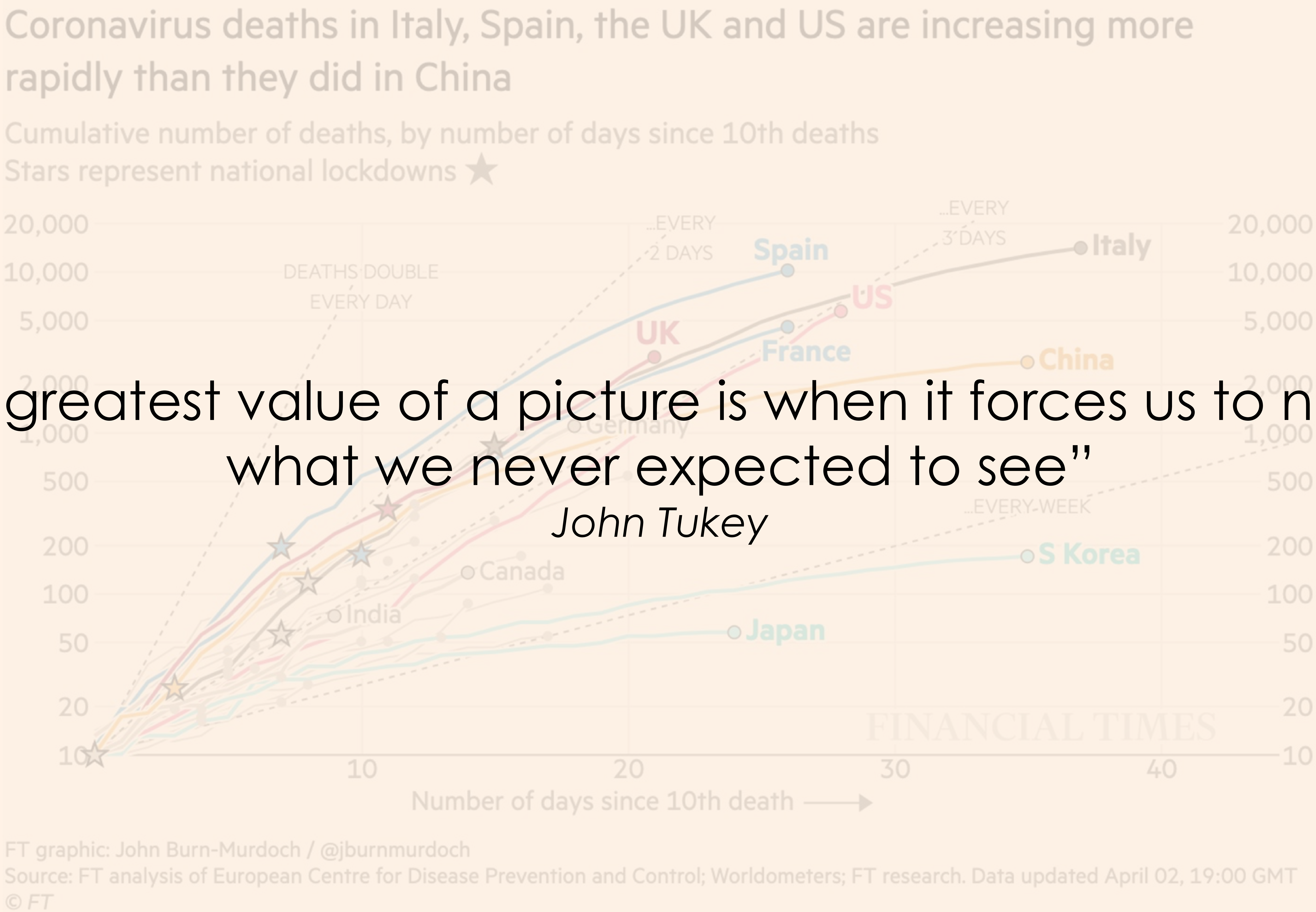
Coronavirus deaths in Italy, Spain, the UK and US are increasing more rapidly than they did in China

Cumulative number of deaths, by number of days since 10th deaths
Stars represent national lockdowns ★



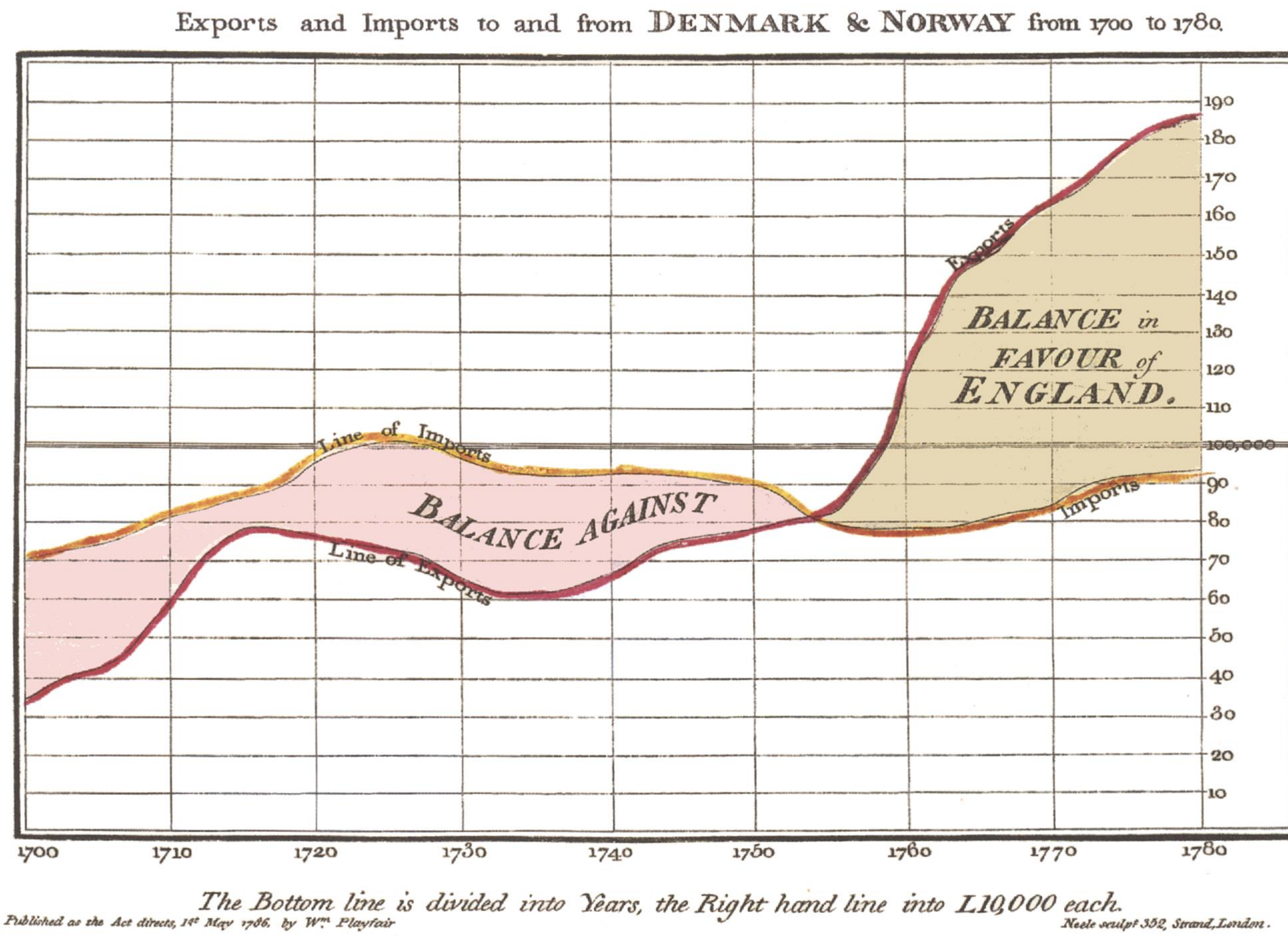
FT graphic: John Burn-Murdoch / @jburnmurdoch
Source: FT analysis of European Centre for Disease Prevention and Control; Worldometers; FT research. Data updated April 02, 19:00 GMT



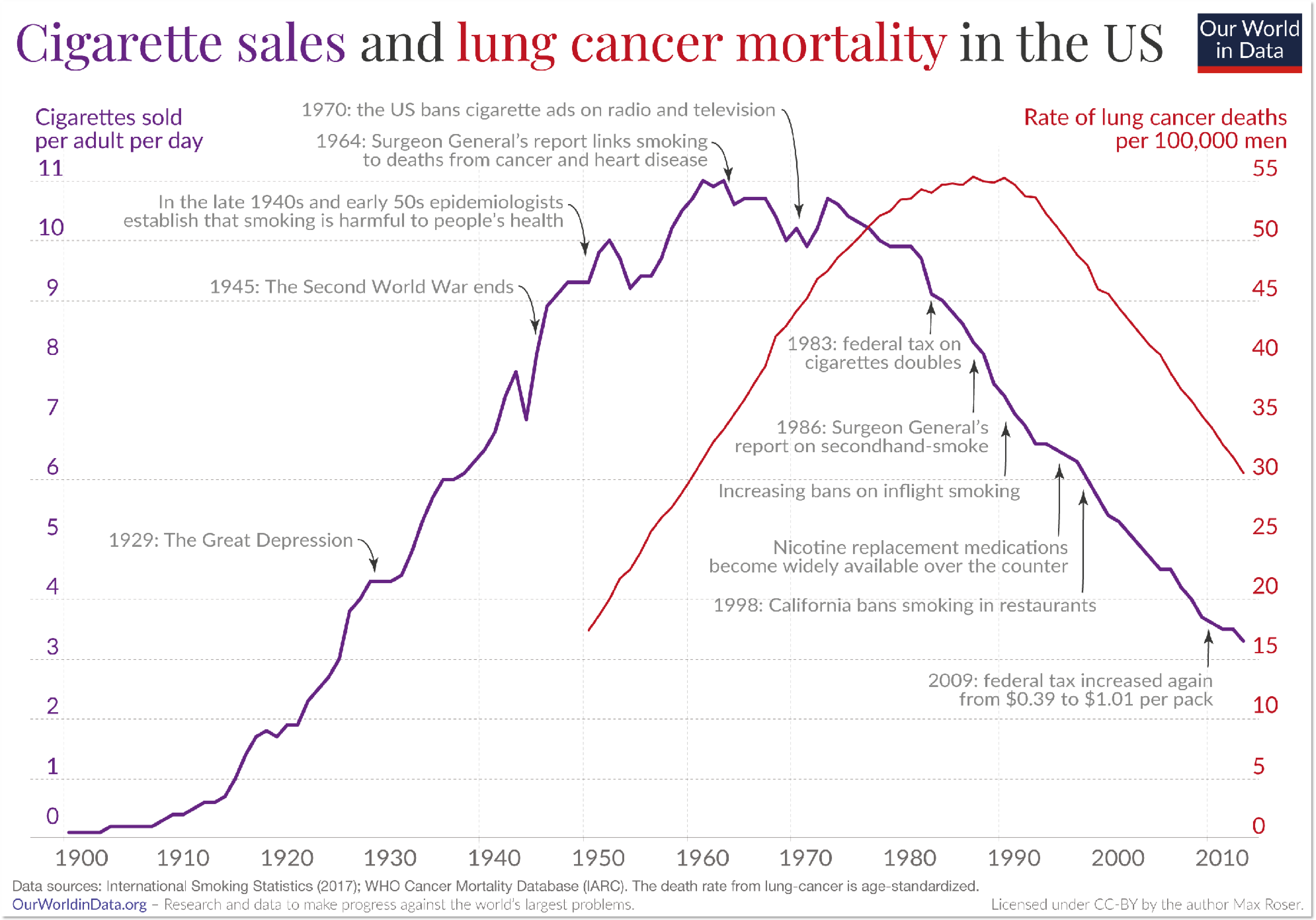
"The greatest value of a picture is when it forces us to notice what we never expected to see"

John Tukey

Data visualization – We tell a story using data



https://upload.wikimedia.org/wikipedia/commons/5/52/Playfair_TimeSeries-2.png


Cigarette sales and lung cancer mortality in the US
Our World in Data
Cigarettes sold per adult per day
Rate of lung cancer deaths per 100,000 men
1970: the US bans cigarette ads on radio and television
1964: Surgeon General's report links smoking to deaths from cancer and heart disease
In the late 1940s and early 50s epidemiologists establish that smoking is harmful to people's health
1945: The Second World War ends
1929: The Great Depression
1983: federal tax on cigarettes doubles
1986: Surgeon General's report on secondhand-smoke
Increasing bans on inflight smoking
Nicotine replacement medications become widely available over the counter
1998: California bans smoking in restaurants
2009: federal tax increased again from $0.39 to $1.01 per pack
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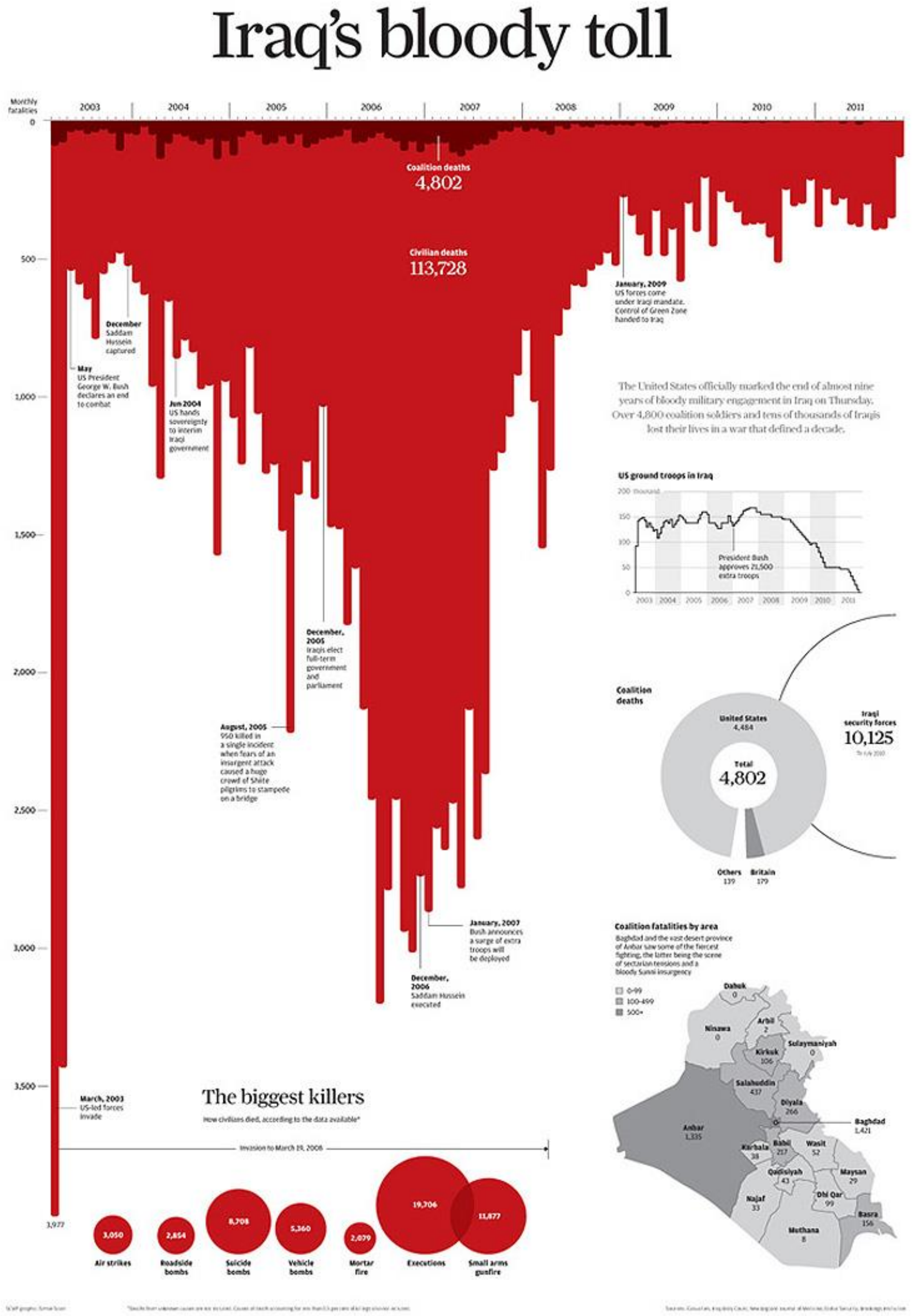
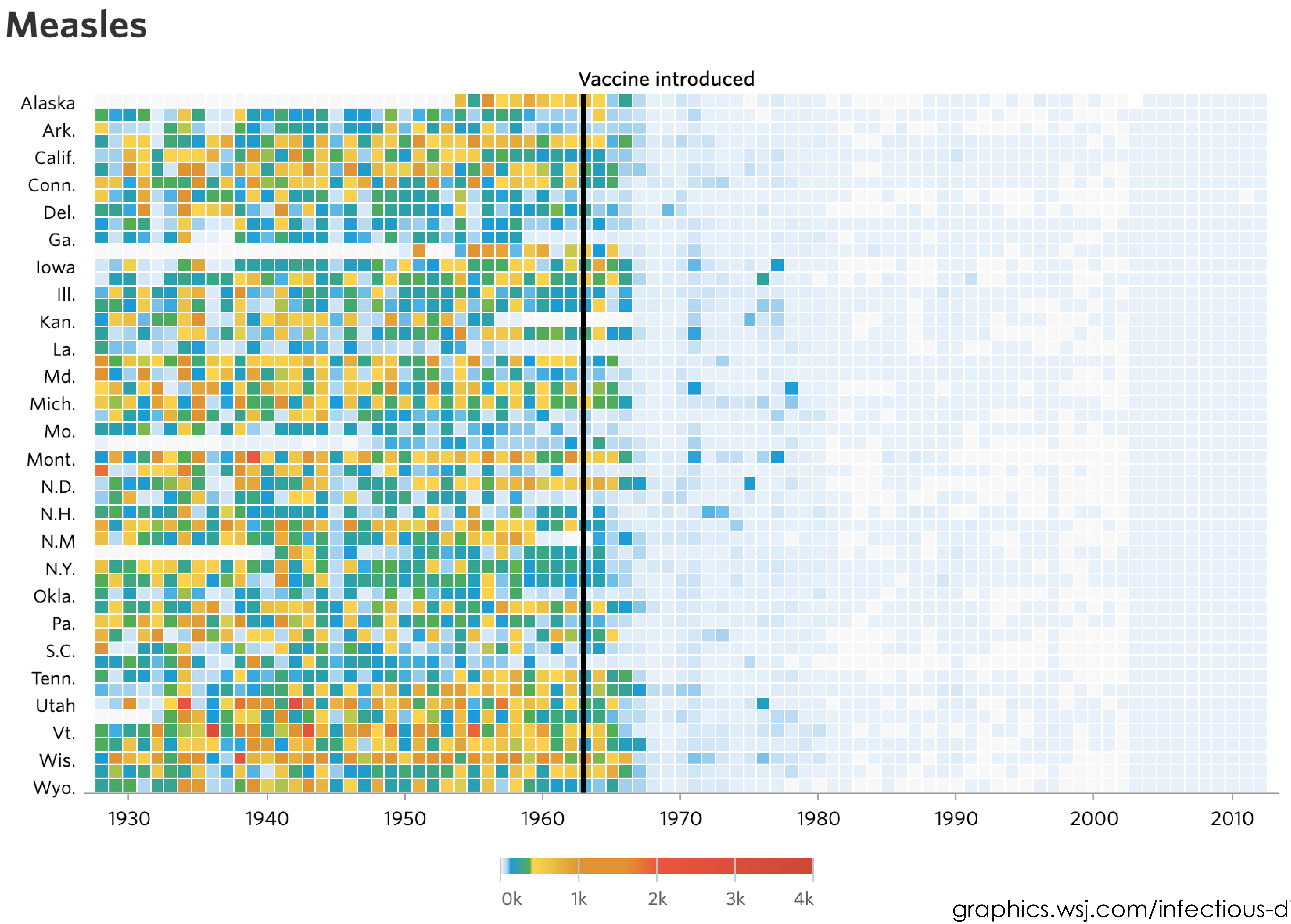
2010
Data sources: International Smoking Statistics (2017); WHO Cancer Mortality Database (IARC). The death rate from lung-cancer is age-standardized.
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Licensed under CC-BY by the author Max Roser.



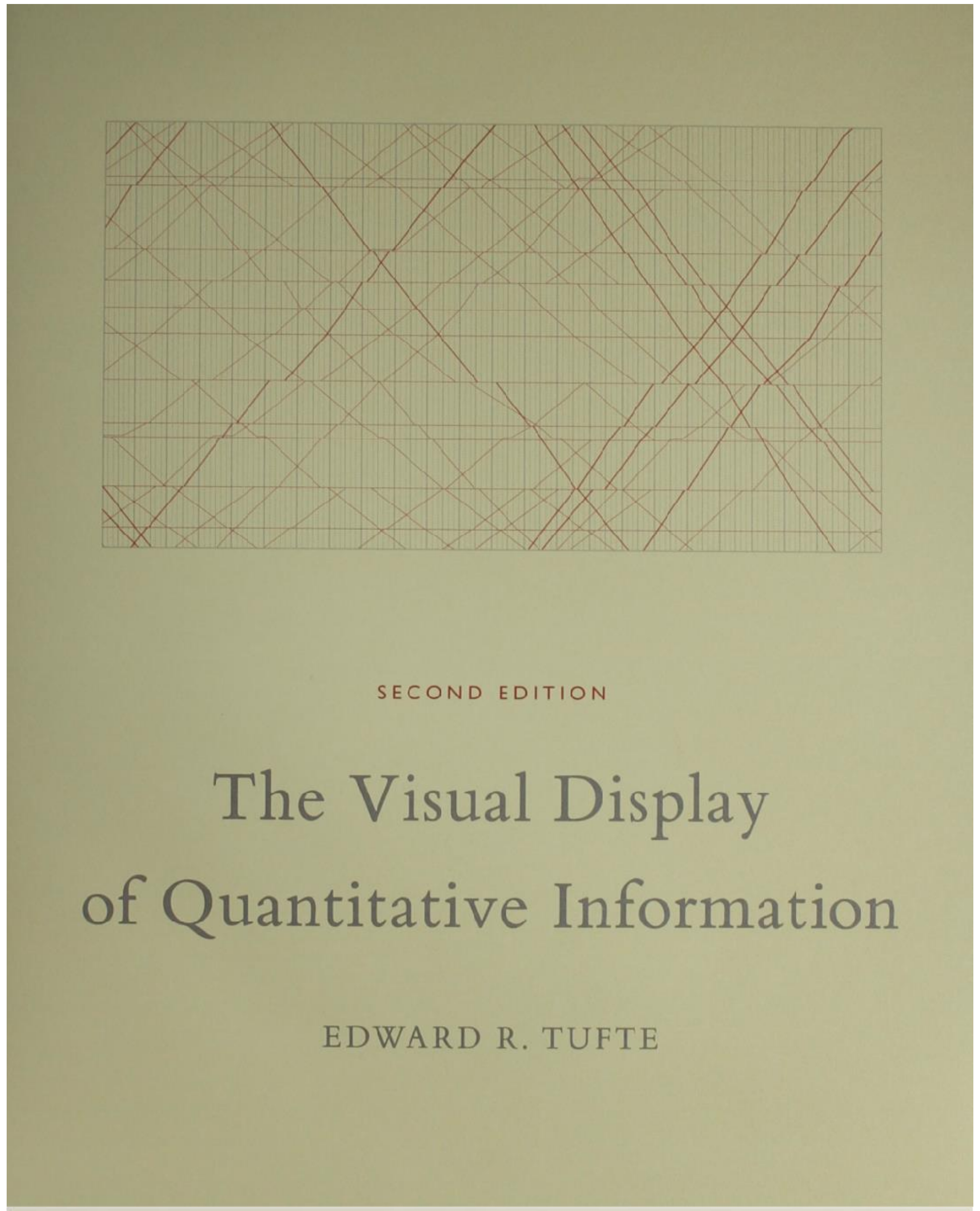
www.simonscarr.com/iraqs-bloody-toll

The impact of vaccines on infectious diseases



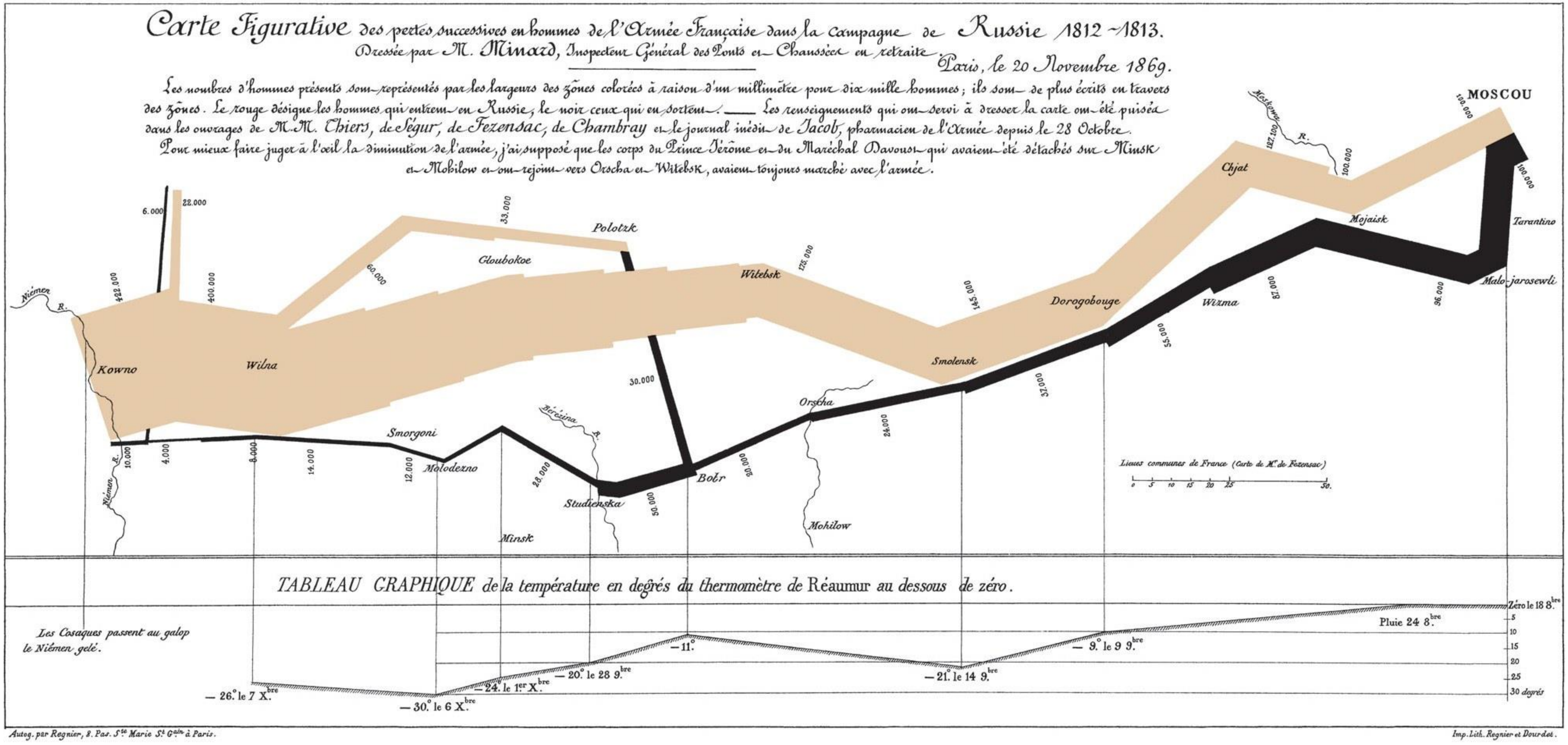
graphics.wsj.com/infectious-diseases-and-vaccines/

Pioneers of data visualization - Edward Tufte ("ET")



www.edwardtufte.com

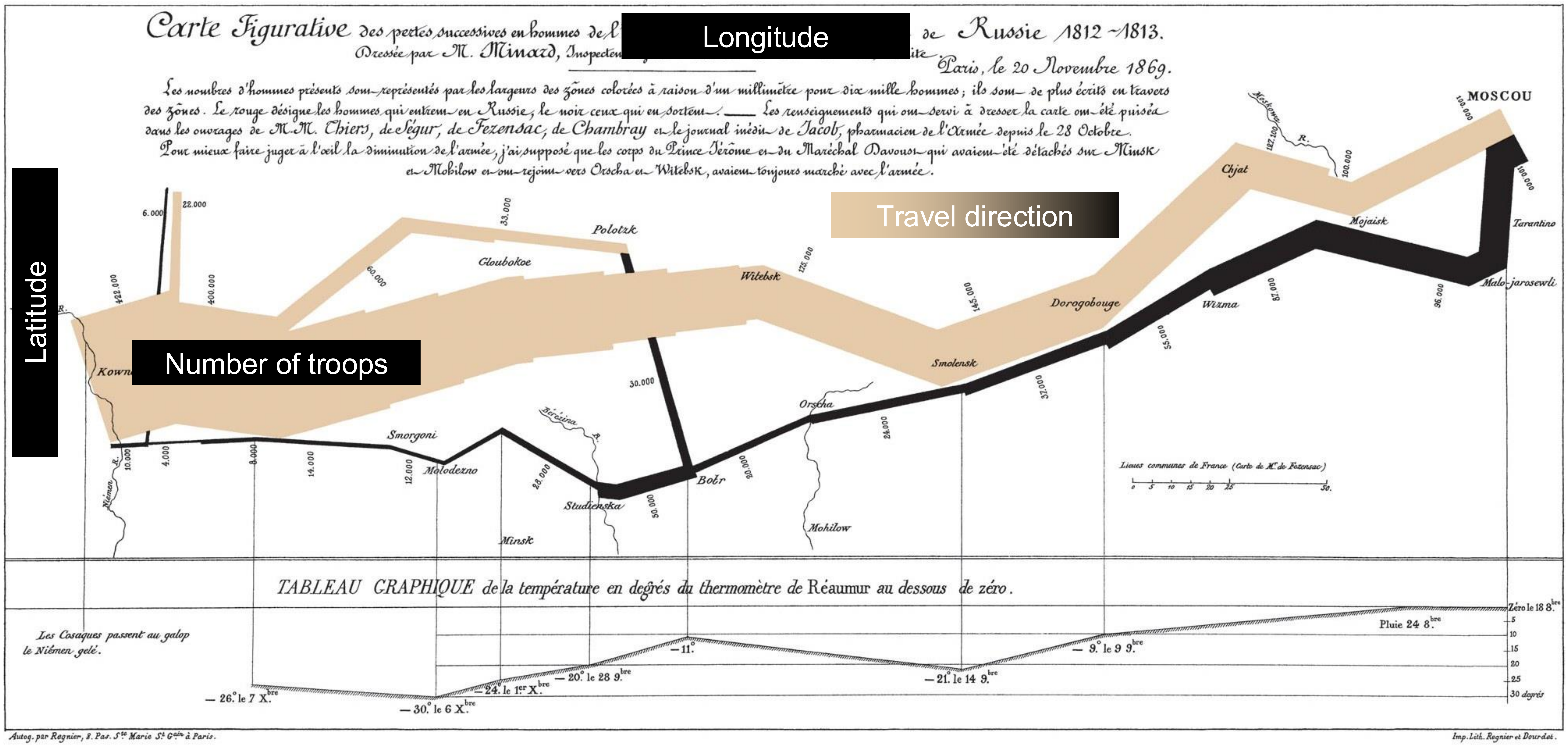
“The best statistical graph ever drawn”



Charles Joseph Minard

“Carte figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812–1813”

“The best statistical graph ever drawn”



Charles Joseph Minard

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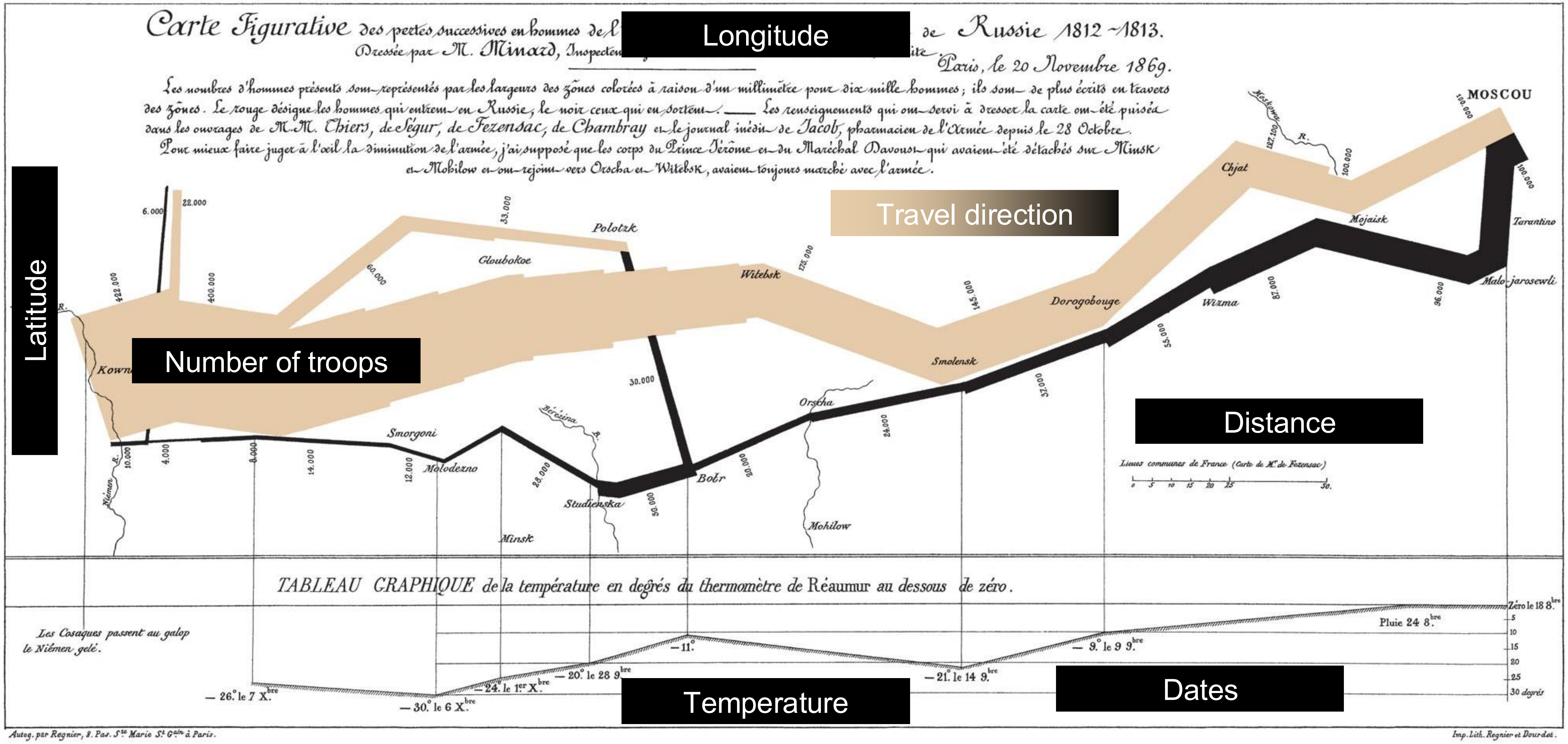
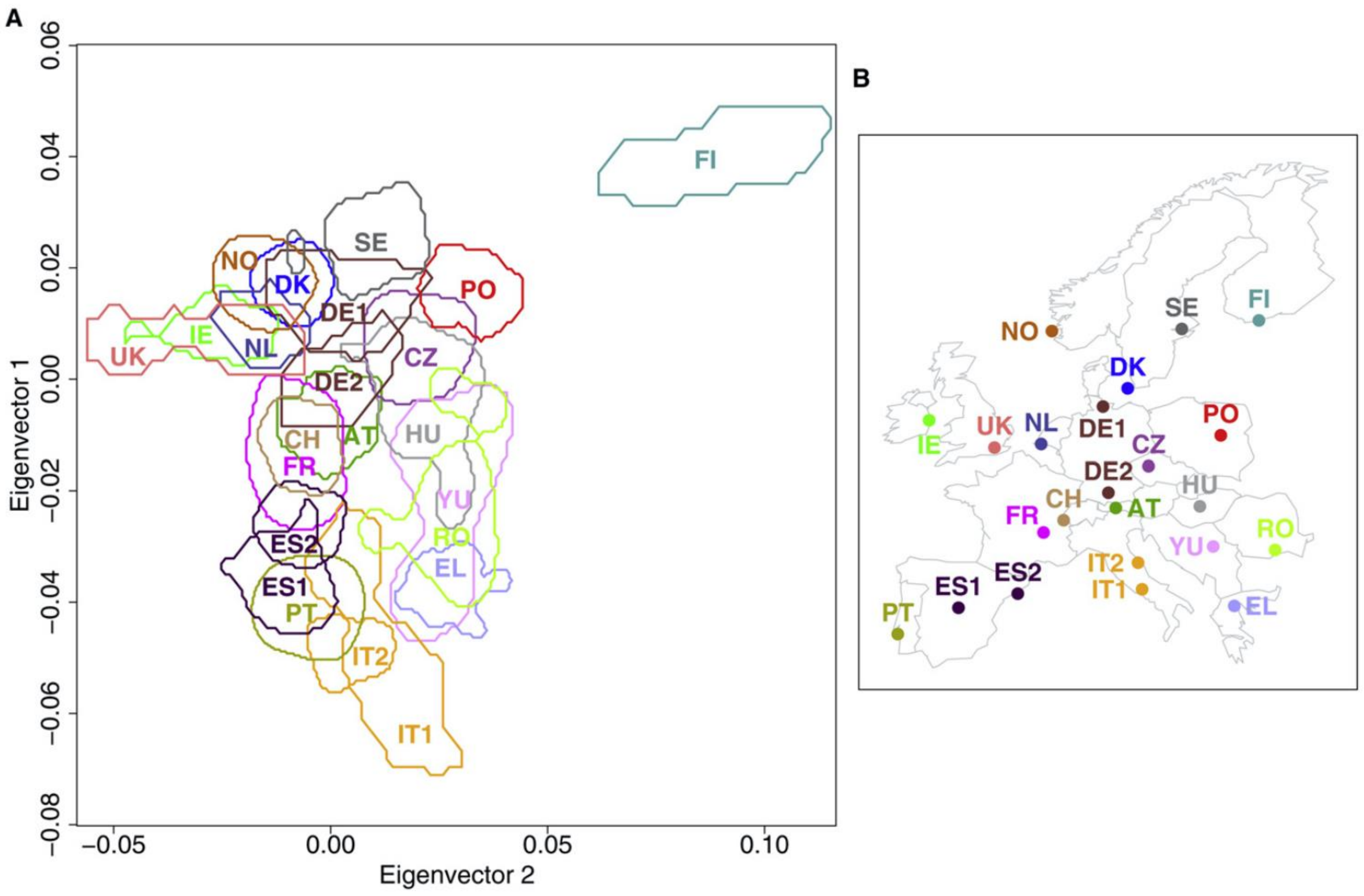


Charles Joseph Minard

“Carte figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812–1813”

A tale of two visualizations

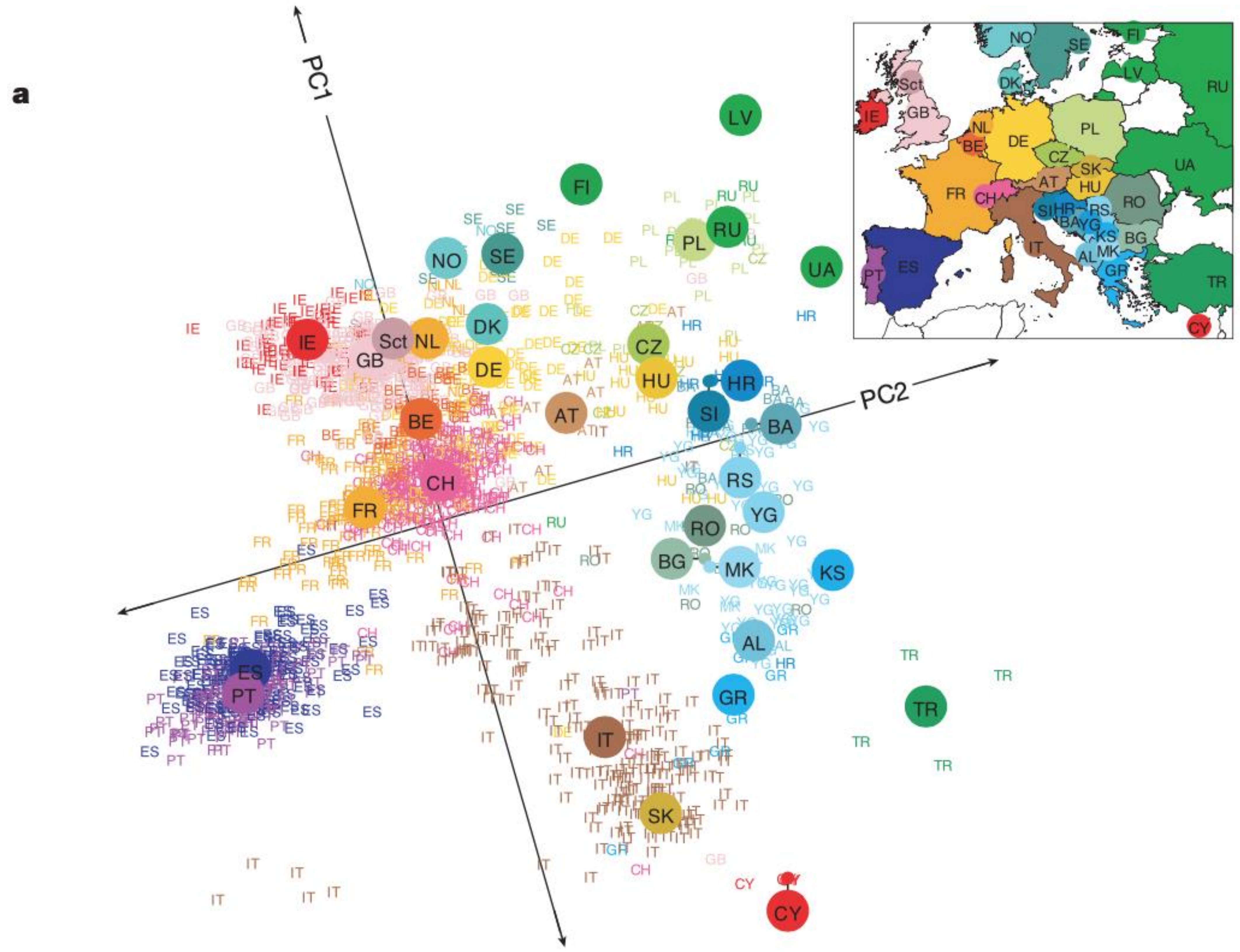
Lao et al 2008



What is the takeaway of this visualization?

A tale of two visualizations

Novembre et al 2008



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A tale of two visualizations

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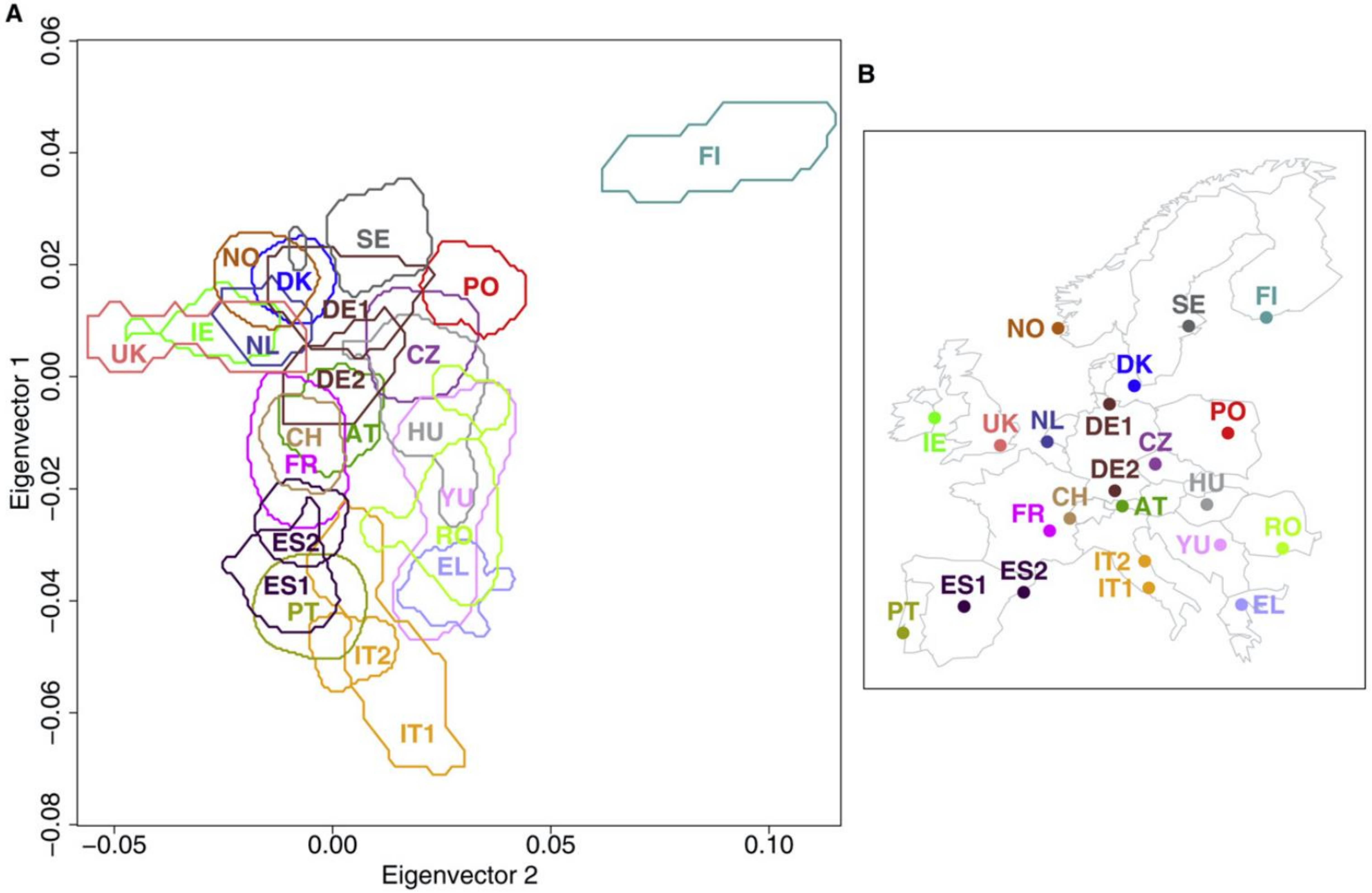
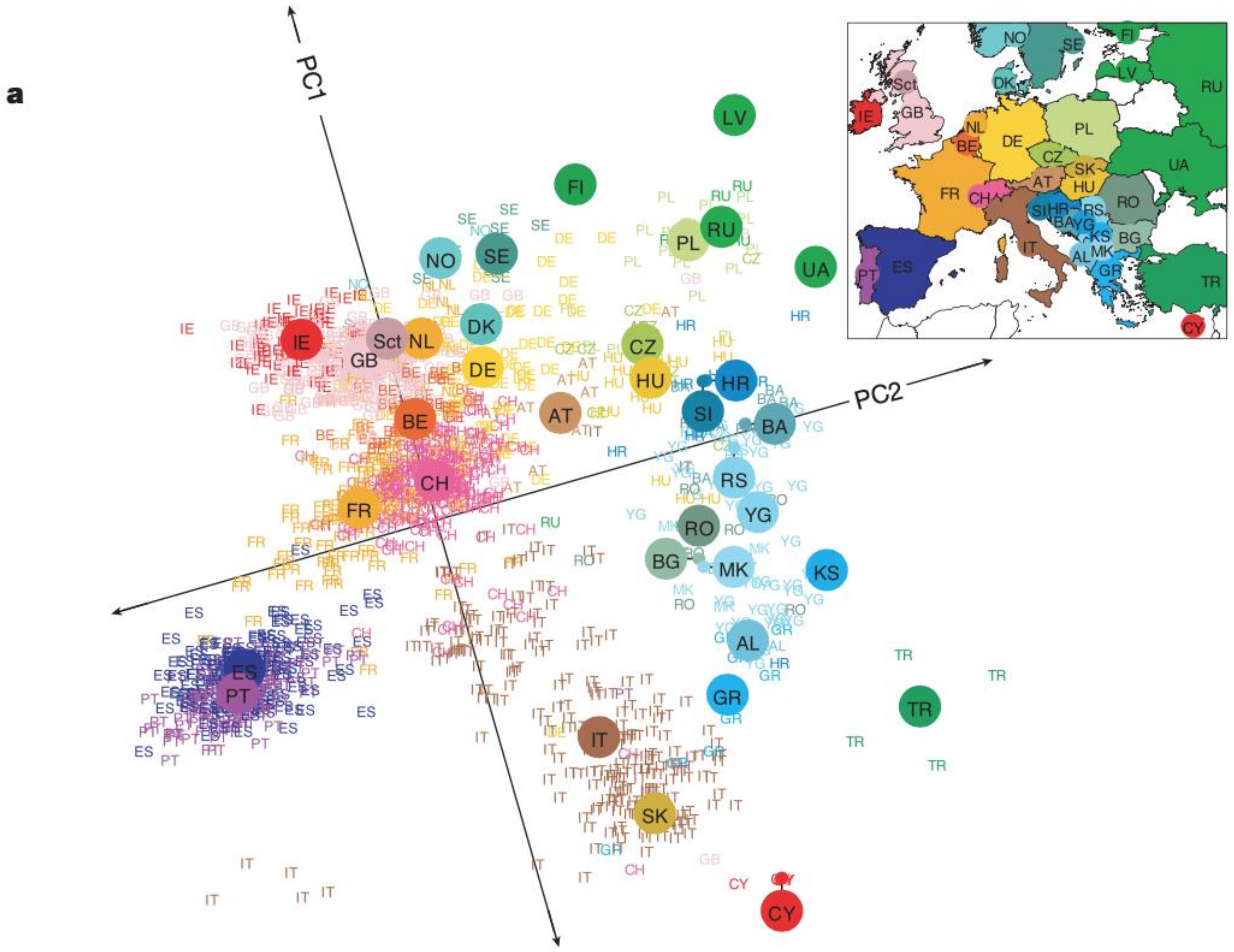


Novembre et al 2008

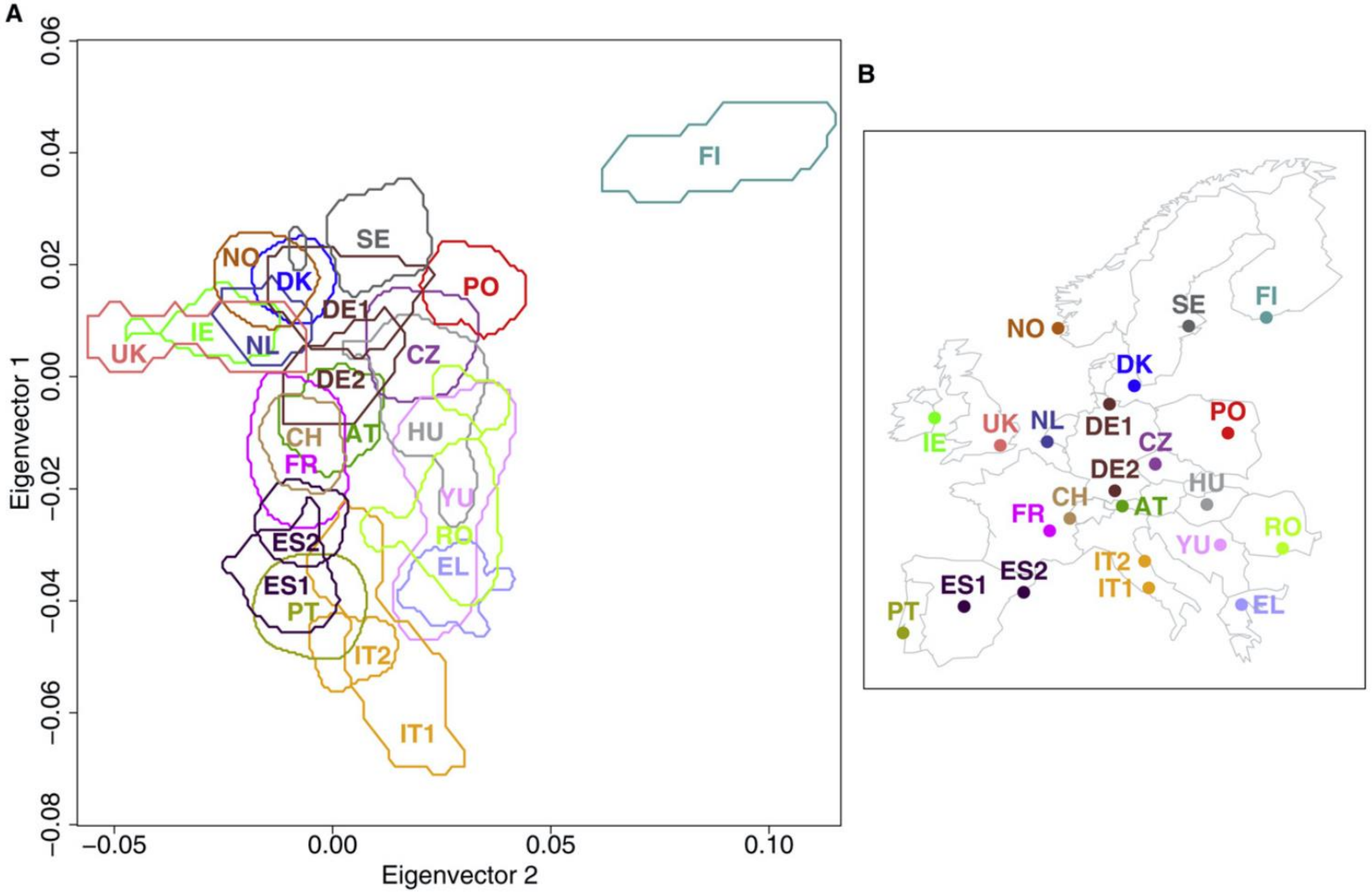


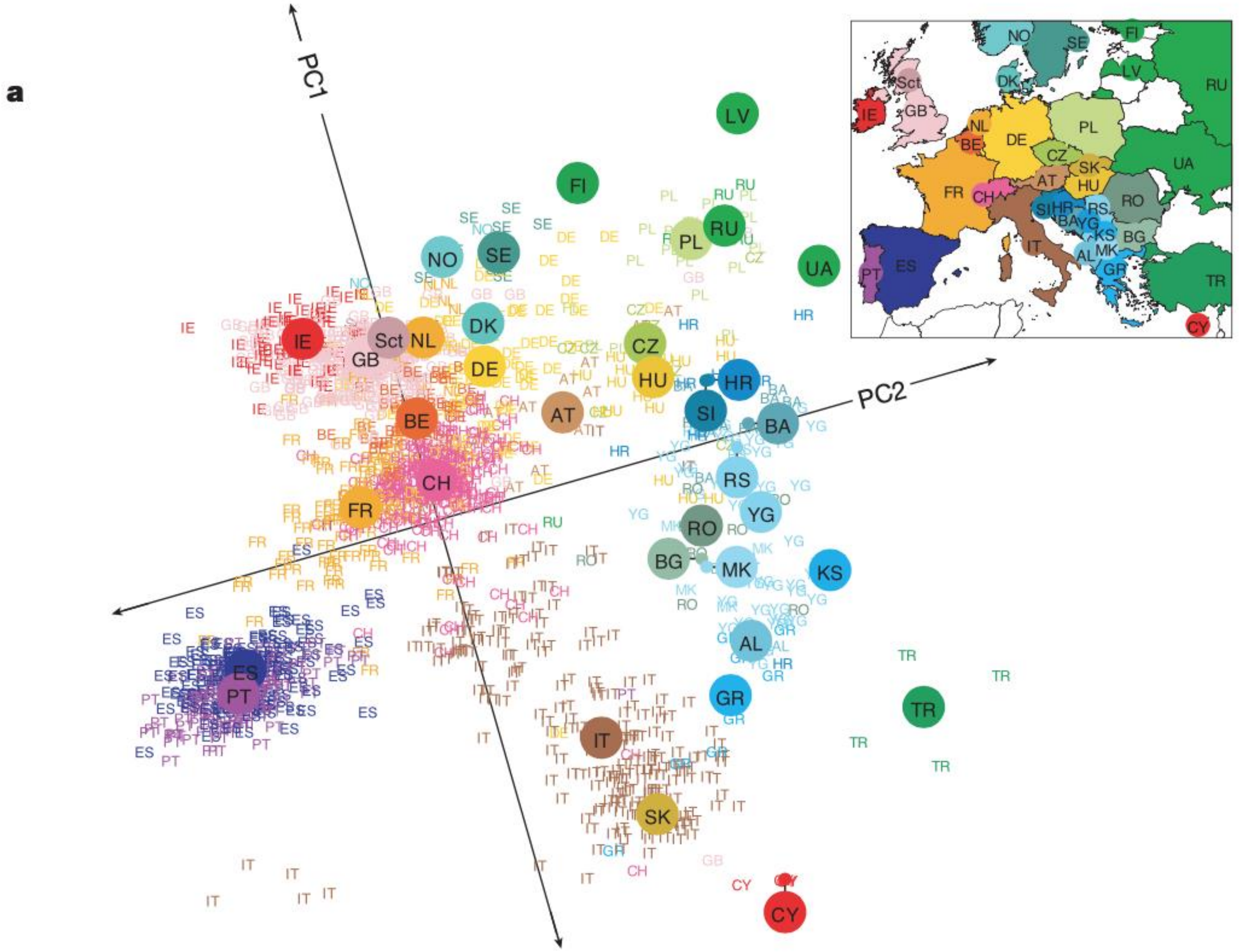
Which one do you prefer?

A tale of two visualizations

Lao et al 2008
"Correlation between Genetic and Geographic Structure in Europe"

Novembre et al 2008
"Genes mirror geography in Europe"

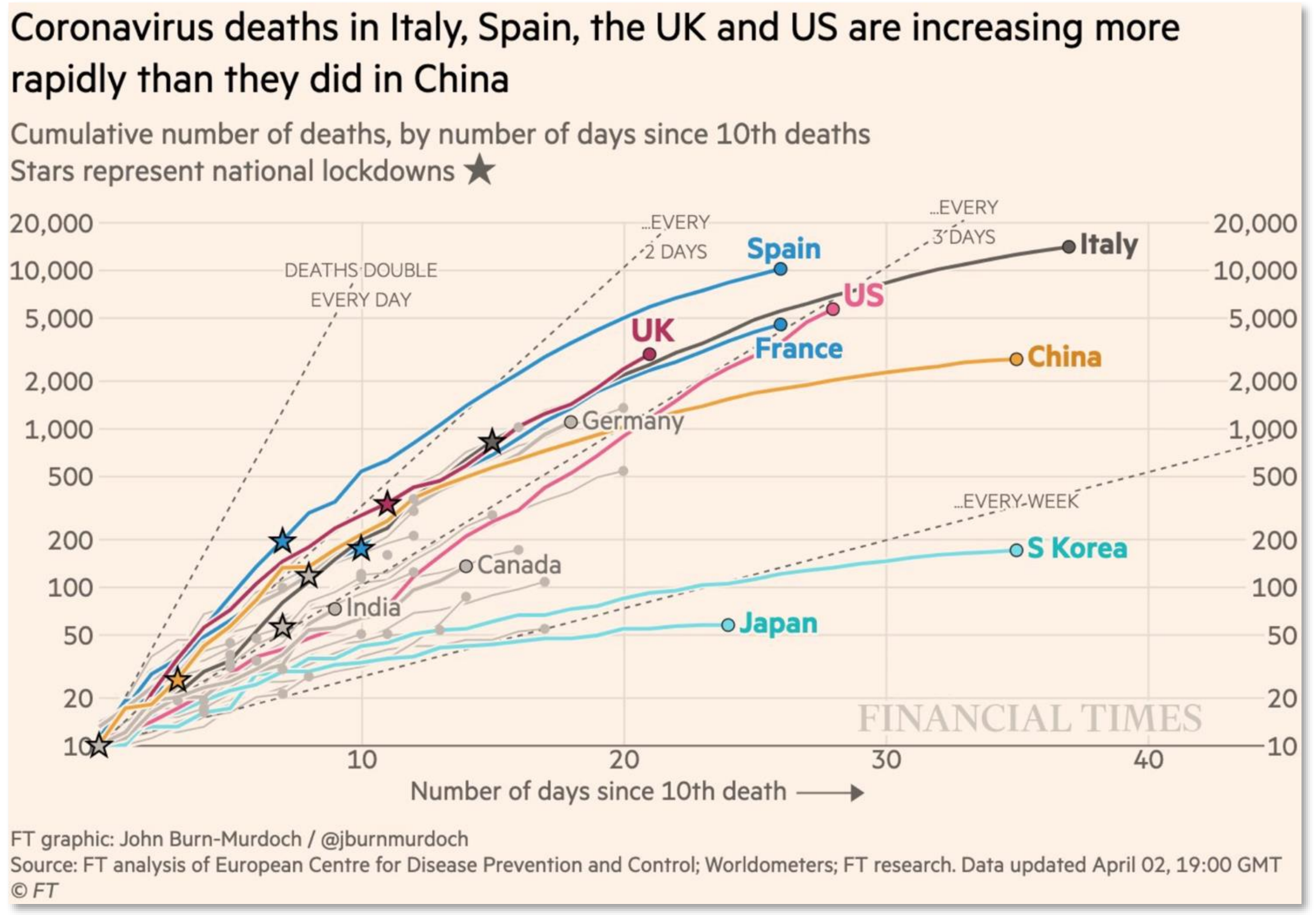




687 citations (Current Biology)

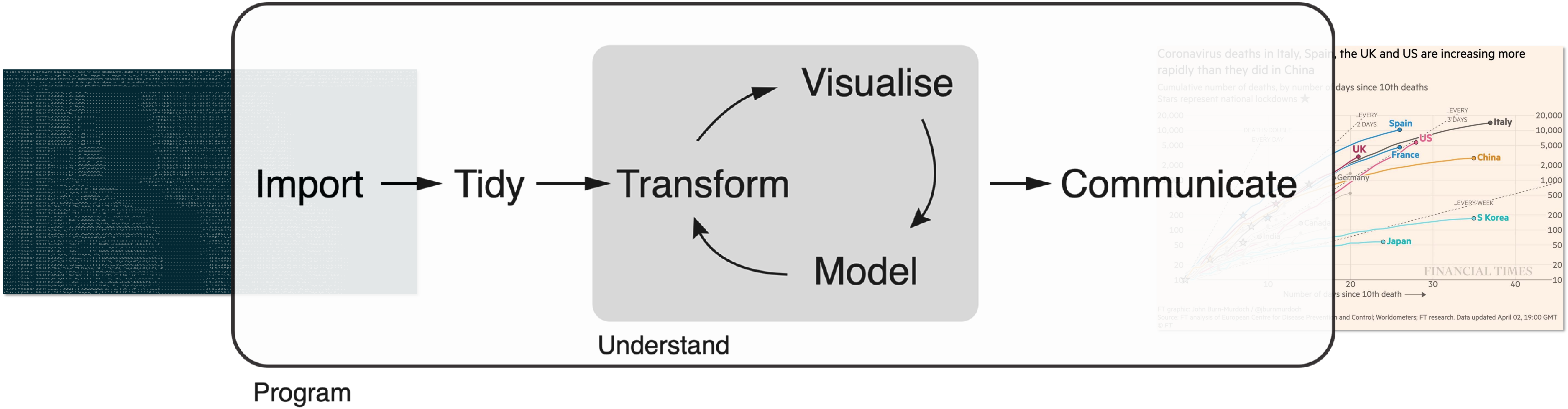
2,111 citations (Nature)

A data science project workflow



https://r4ds.had.co.nz/introduction.html

A data science project workflow



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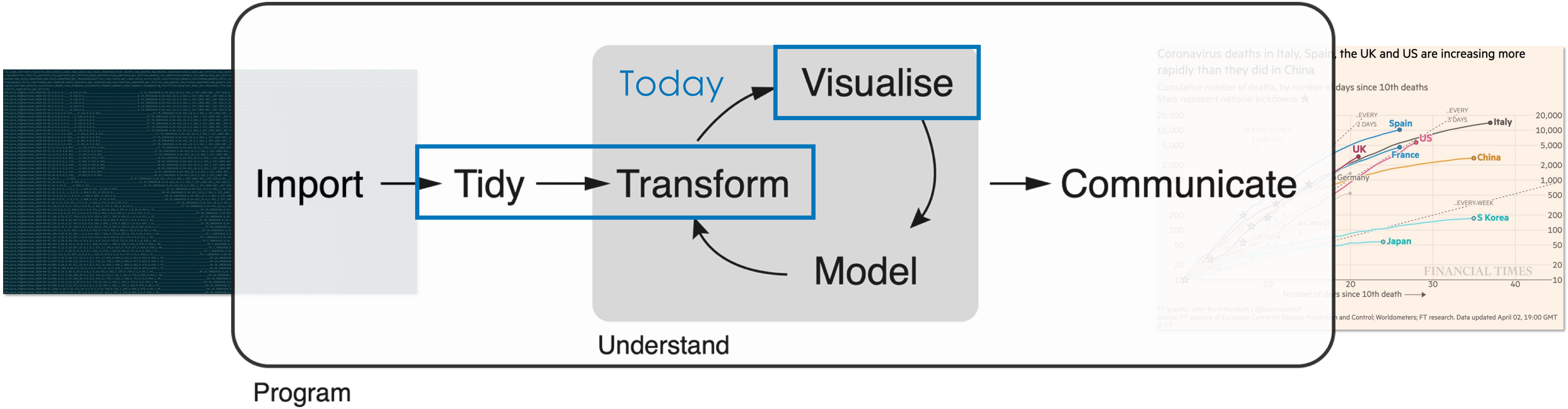
A data science project workflow



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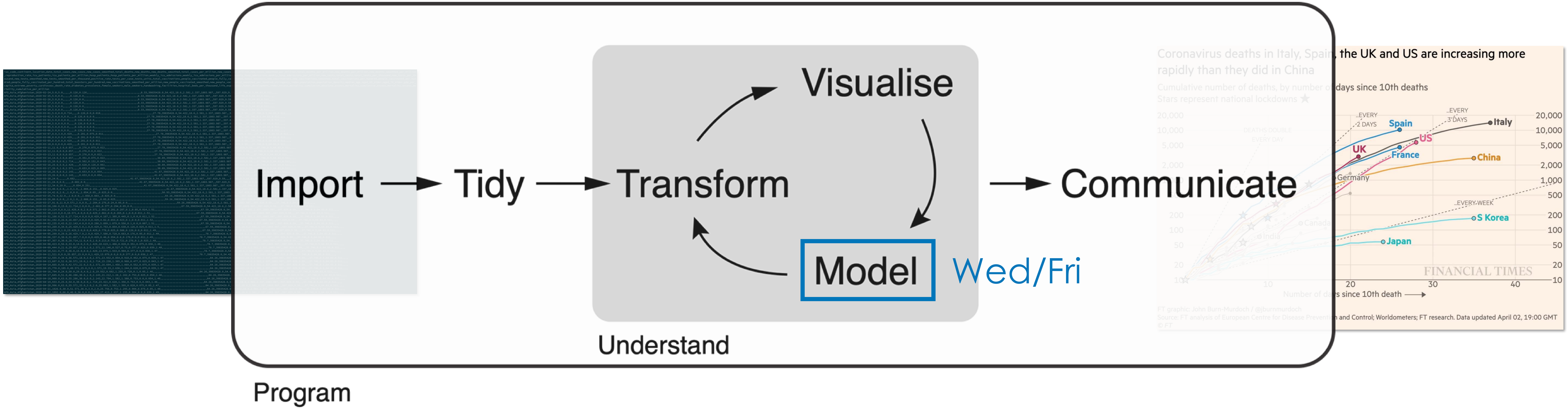
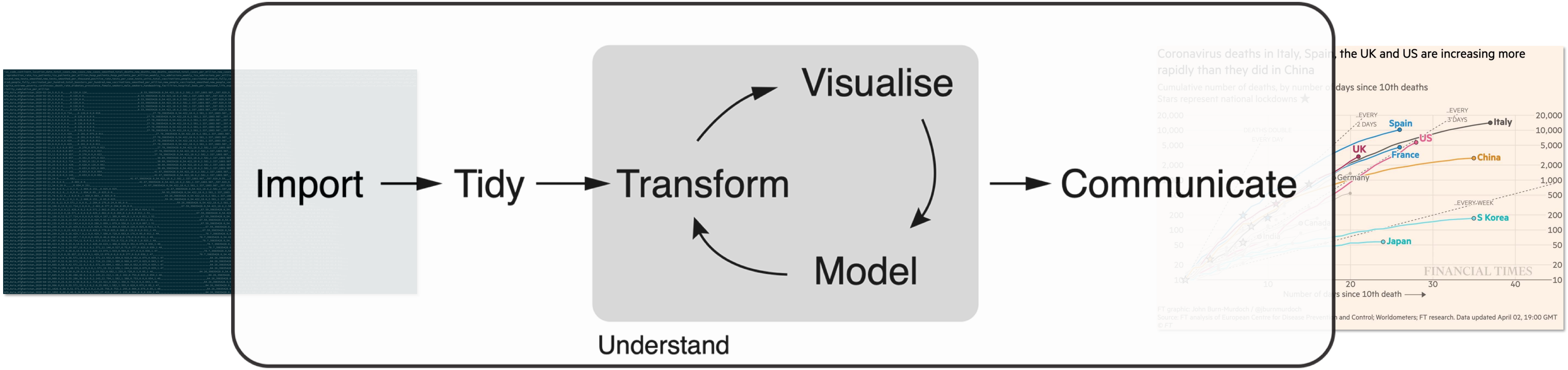
A data science project workflow



https://r4ds.had.co.nz/introduction.html

A data science project workflow



Program

Week 2 Mon / Tue

https://r4ds.had.co.nz/introduction.html

When data is messy …

species	habitat	weight	length	latitude/longitude	date
Alligator mississippiensis	swamp	431 lb	4 ft 2	29.531,-82.184	Sept 15, 2015
Puma concolor	forest	125 lb	2.2m	29.125,-81.682	08/10/2015
Ursus americanus	forest	88 kg	133 cm	N29°7'30"/W81°40'55.2"	07-13-2015

In which ways is this dataset "untidy"?

Hart et al (2016) PLoS Comp Biol (https://doi.org/10.1371/journal.pcbi.10050

When data is messy … we tidy it up

species	habitat	weight	length	latitude/longitude	date
Alligator mississippiensis	swamp	431 lb	4 ft 2	29.531,-82.184	Sept 15, 2015
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Ursus americanus	forest	88 kg	133 cm	N29°7'30"/W81°40'55.2"	07-13-2015

meta-data

data

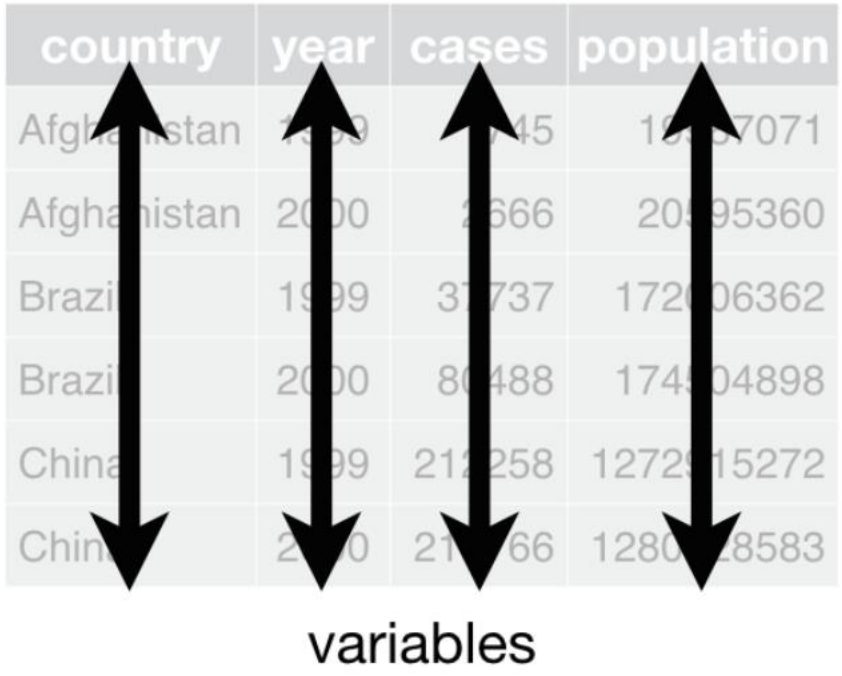
species_code	date	station_code	weight_kg	length_cm
TSN 551771	2015-09-15	1	196	127
TSN 55247	2015-08-10	2	57	220
TSN 180544	2015-07-13	2	88	133

station_code	habitat	latitude	longitude
1	swamp	29.531	-82.184
2	forest	29.125	-81.682

species_code	class	genus	species
TSN 551771	Reptilia	Alligator	mississippiensis
TSN 55247	Mammalia	Puma	concolor
TSN 180544	Mammalia	Ursus	americanus

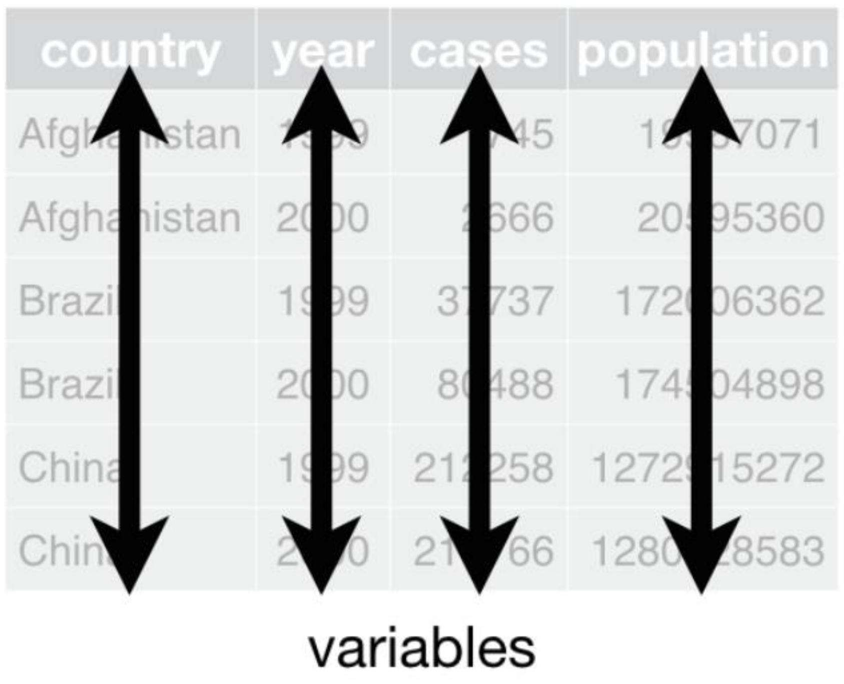
Hart et al (2016) PLoS Comp Biol (https://doi.org/10.1371/journal.pcbi.10050

Principles of tidy data

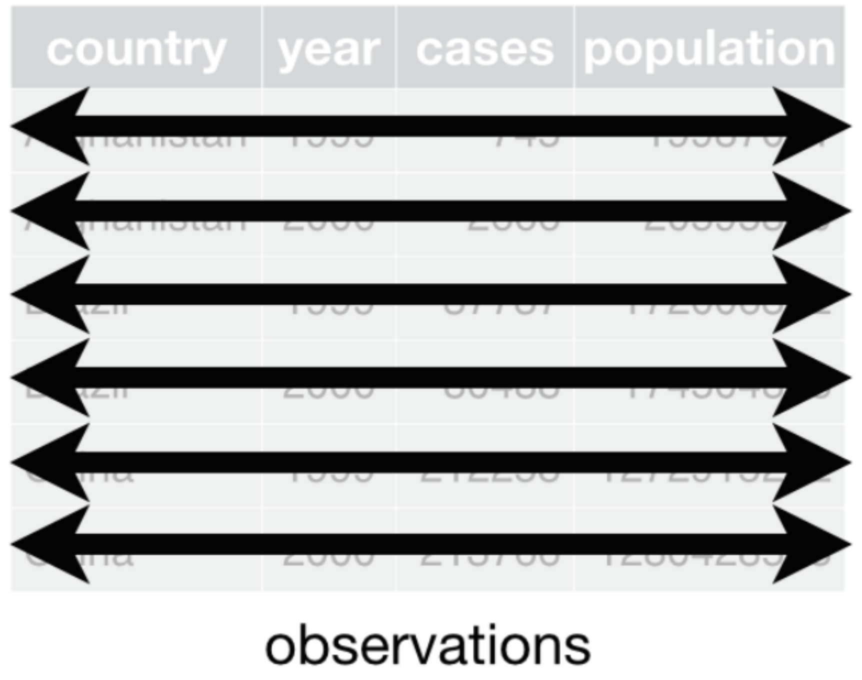


1. Each variable has its own column

Principles of tidy data



1. Each variable has its own column



2. Each observation has its own row

Principles of tidy data

variables

1. Each variable has its own column

observations

2. Each observation has its own row

values

3. Each values has its own cell

What does one row represent?

country	date	new_cases	new_deaths	population
Denmark	2020-02-28	1	0	5,902,901
Denmark	2020-03-01	2	0	5,902,901
Denmark	2020-03-02	1	0	5,902,901
India	2020-01-30	1	0	1,425,423,209
India	2020-02-02	1	0	1,425,423,209
India	2020-02-03	1	0	1,425,423,209

2,394
days per country

32
countries

76,544
rows

data/covid_day2.csv — OWID COVID extract to 2026-07-19

What does one row represent?

country	date	new_cases	new_deaths	population
Denmark	2020-02-28	1	0	5,902,901
Denmark	2020-03-01	2	0	5,902,901
Denmark	2020-03-02	1	0	5,902,901
India	2020-01-30	1	0	1,425,423,209
India	2020-02-02	1	0	1,425,423,209
India	2020-02-03	1	0	1,425,423,209

One country, one day

- mean(new_cases) averages country-days, not countries
- India and Denmark contribute one row each per day — 1.4 billion people weighted the same as 5.9 million
- A country that reported for two years counts twice what one that reported for one does

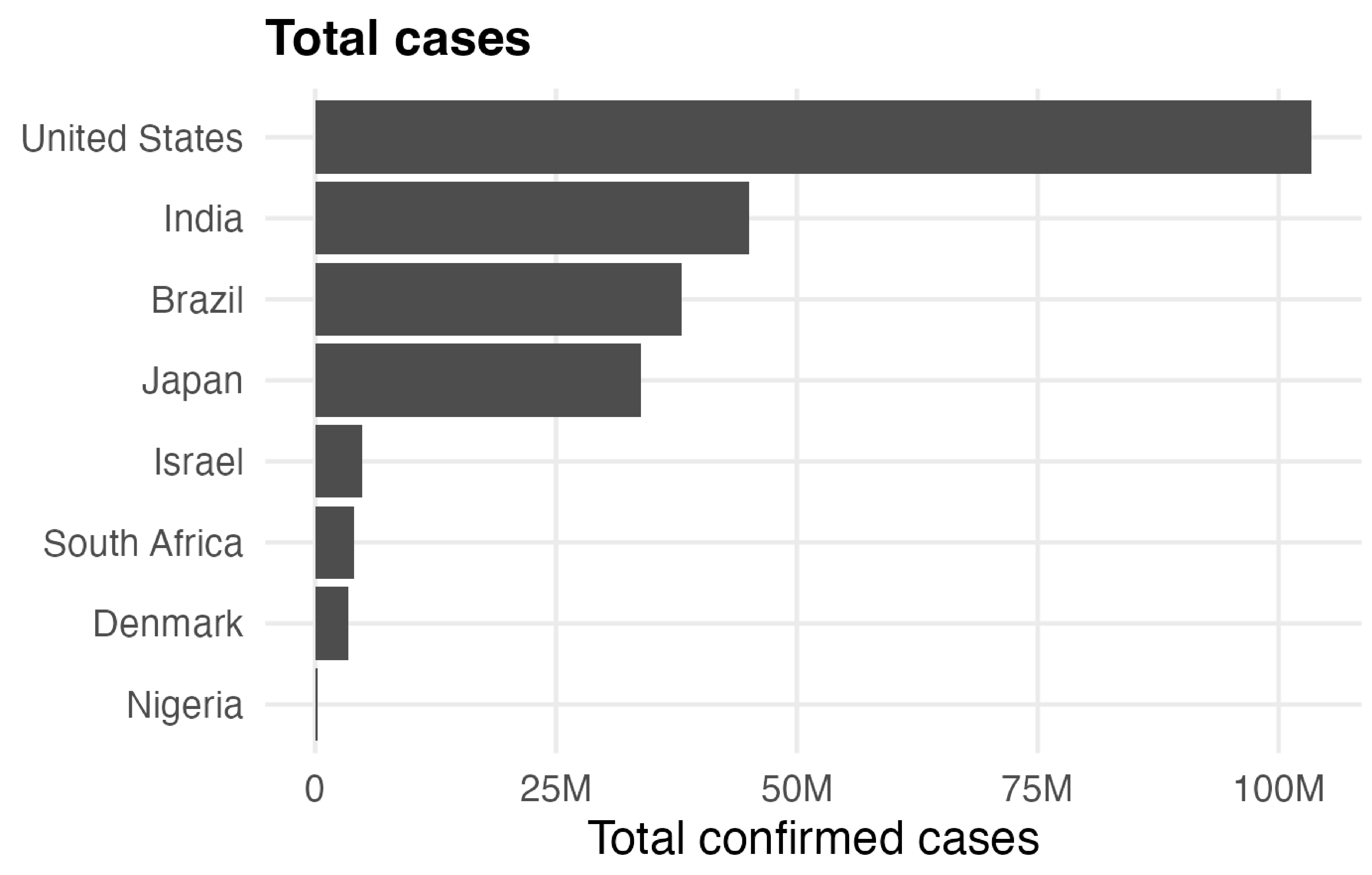
2,394
days per country

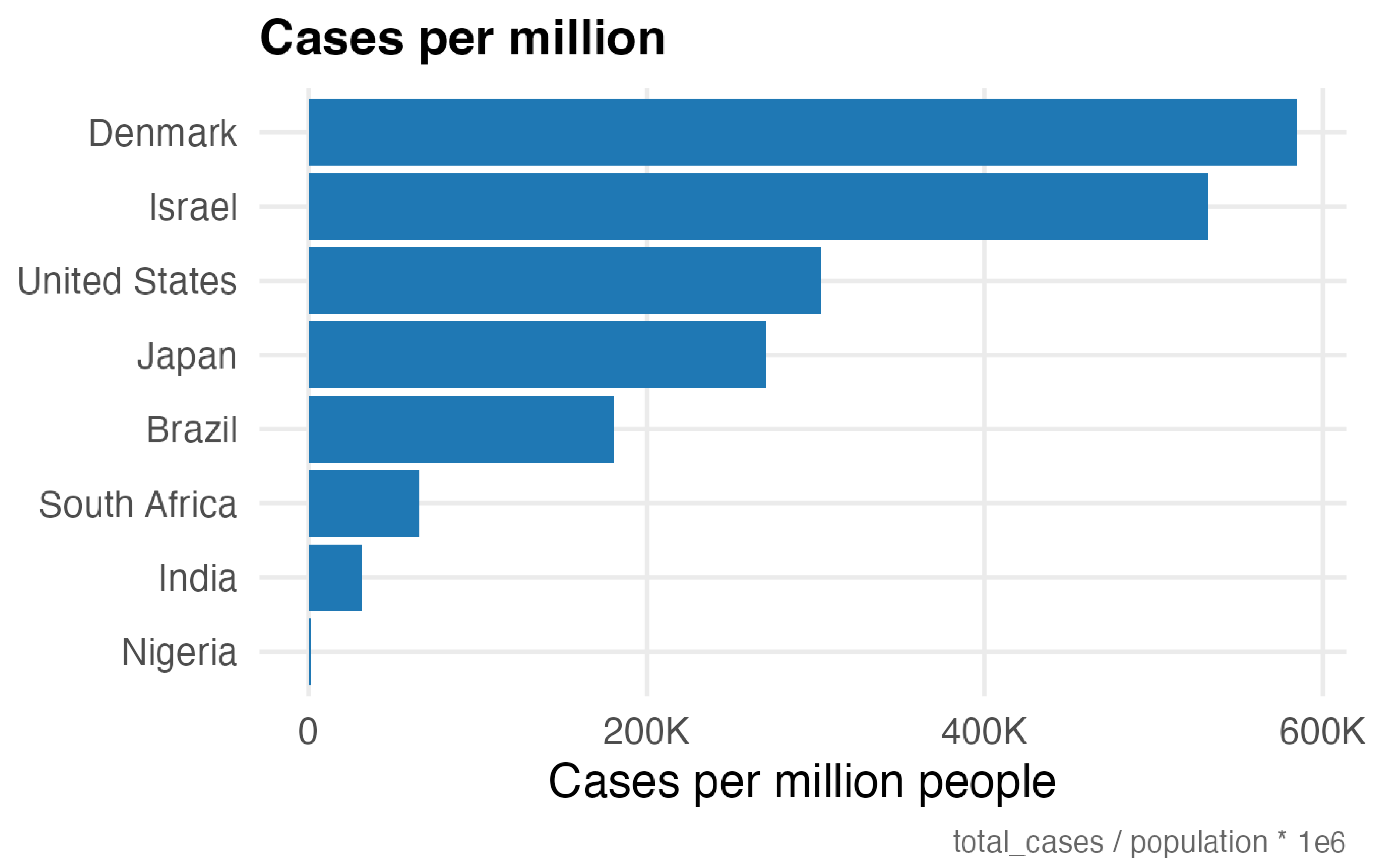
32
countries

76,544
rows

data/covid_day2.csv — OWID COVID extract to 2026-07-19

10,000 cases is not a number





Which country was hit hardest?

- India is 2nd by count and 7th by rate
- Denmark is 7th by count and 1st by rate
- Neither chart is wrong. They answer different questions.

total_cases vs total_cases / population * 1e6

Per what?

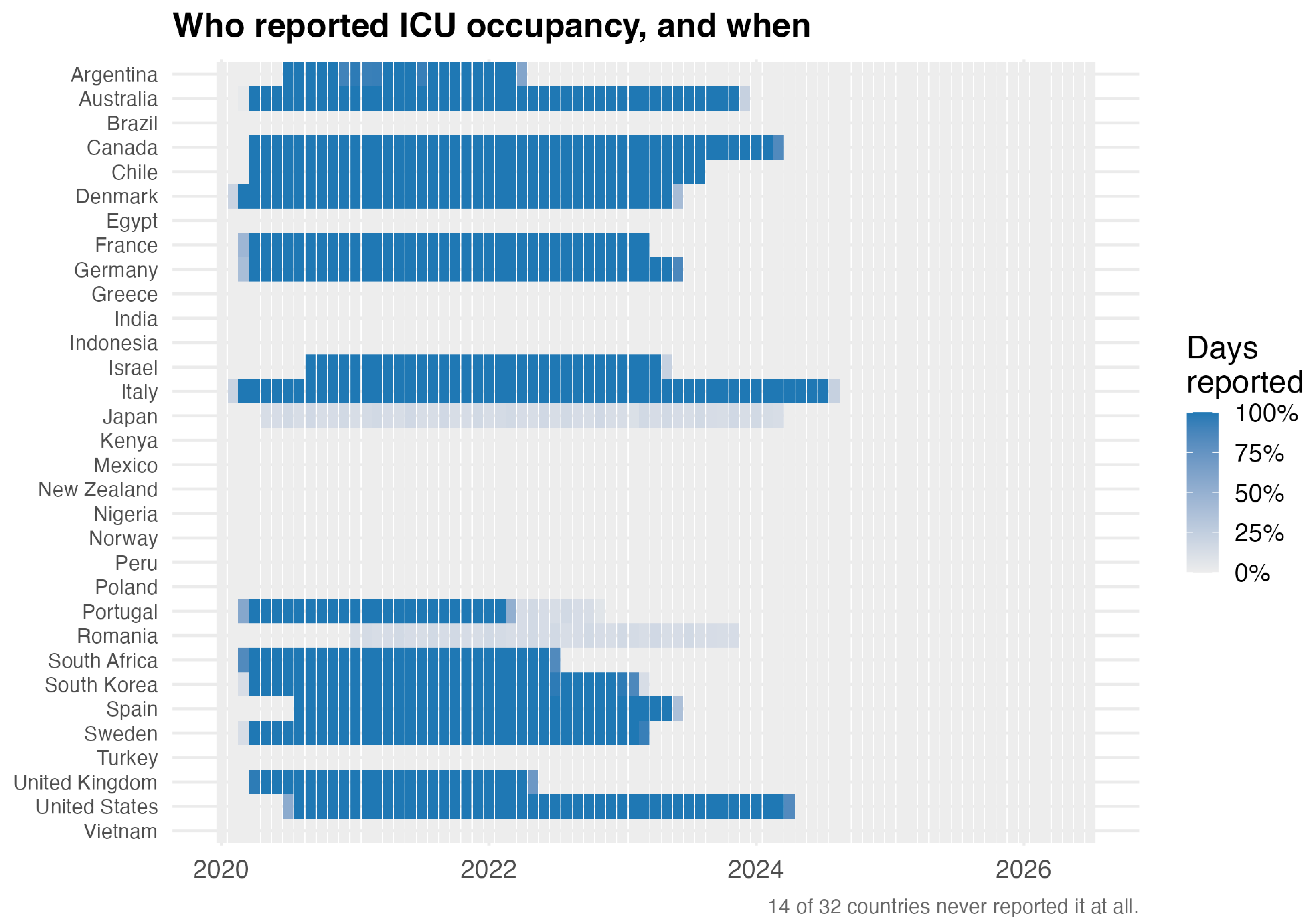
One numerator, three denominators, three different claims

new_deaths / population	mortality rate	how much of the population died
new_deaths / new_cases	case fatality ratio	how dangerous a detected infection was
new_cases / new_tests	positivity	how hard you were looking

A denominator is a measurement too. Testing varied a hundredfold between countries, so the case fatality ratio mostly measures testing policy.

https://padlet.com/ucph/fundamentals_2026_module2_wrangle

Missing data, kind one and two



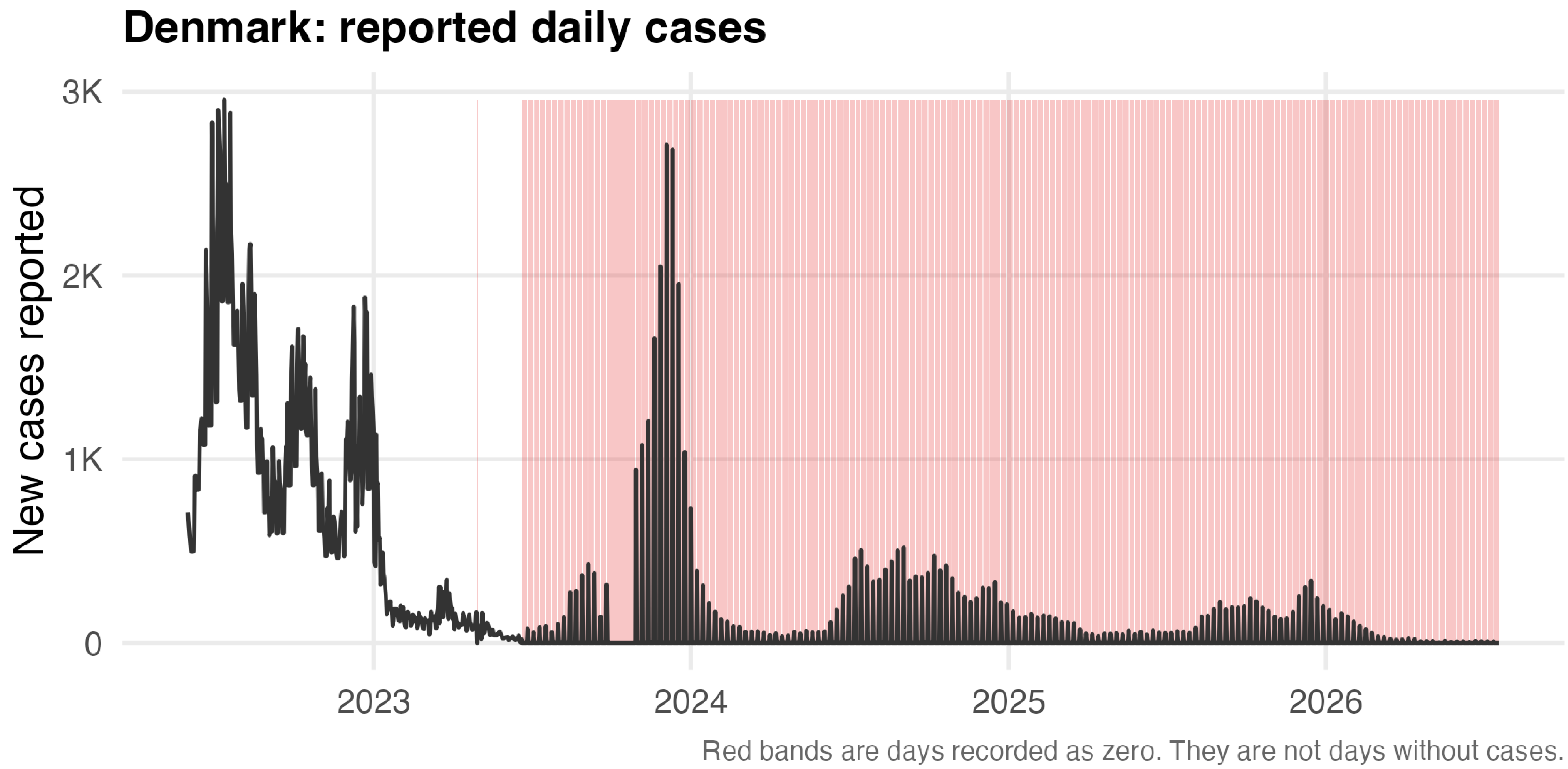
14 of 32

countries never reported ICU occupancy — not once in 2,394 days

- Never collected: an absent variable, not missing data
- Collected for a while: Denmark and Israel stop partway through. The data decides your analysis period

Missing data, kind three

Missing disguised as zero — the dangerous one



Denmark, new_cases

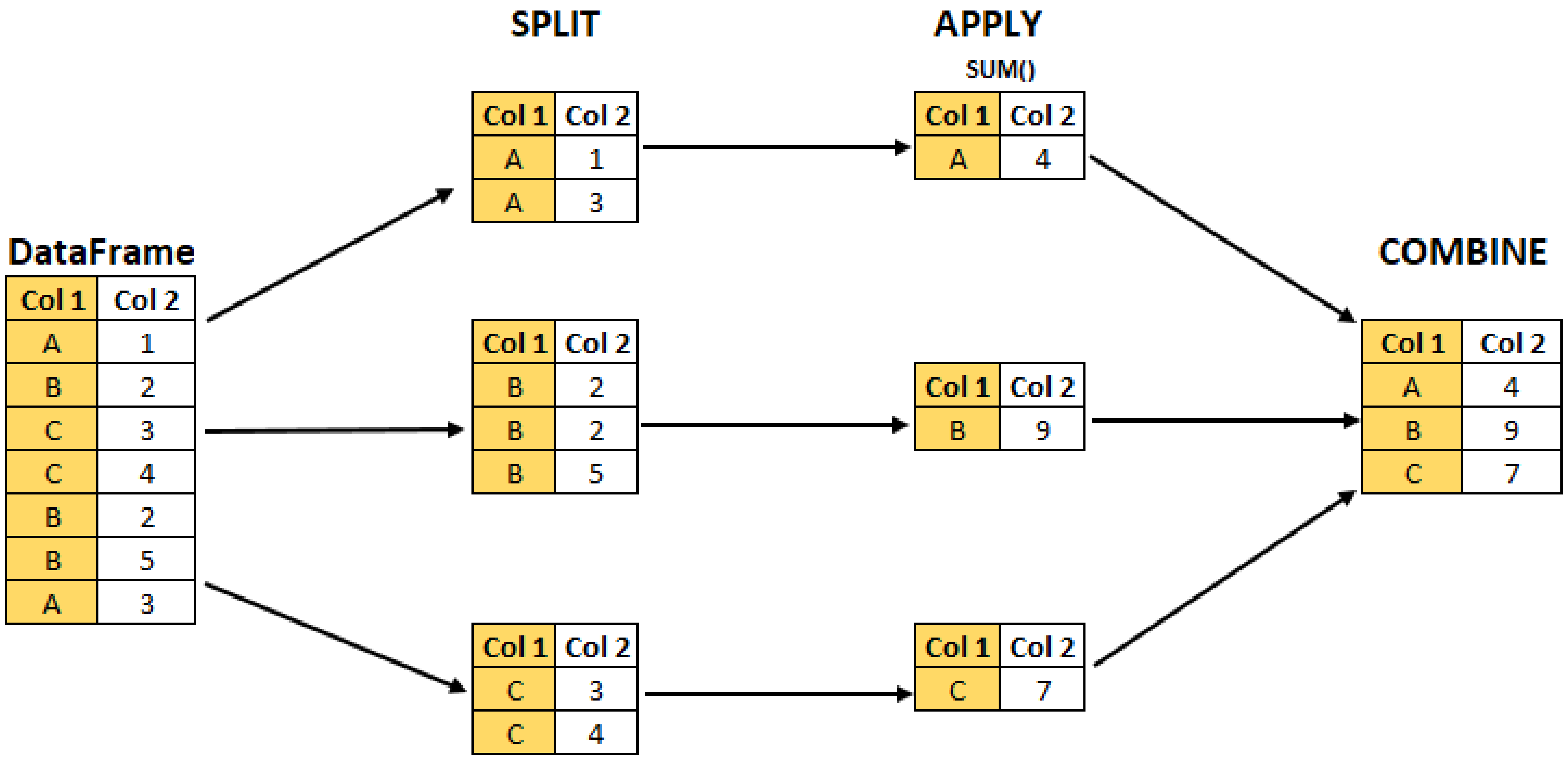
year	days	NA	zero
2021	365	0	0
2022	365	0	0
2023	365	0	172
2024	366	0	313
2025	365	0	313

Zero cases and no report look identical

`filter(!is.na(new_cases))` keeps every one of them

Plot the variable over time before you summarise it. The comb pattern is visible in one second and invisible in every summary statistic.

Data transformation



split-apply-combine

https://medium.com/analytics-vidhya/split-apply-combine-strategy-for-data-mining-4fd6e2a0cc99

Data transformation

```
mtcars  %>%
  group_by(cyl) %>%
  summarise(avg = mean(mpg))
```

Group and summarise

dplyr

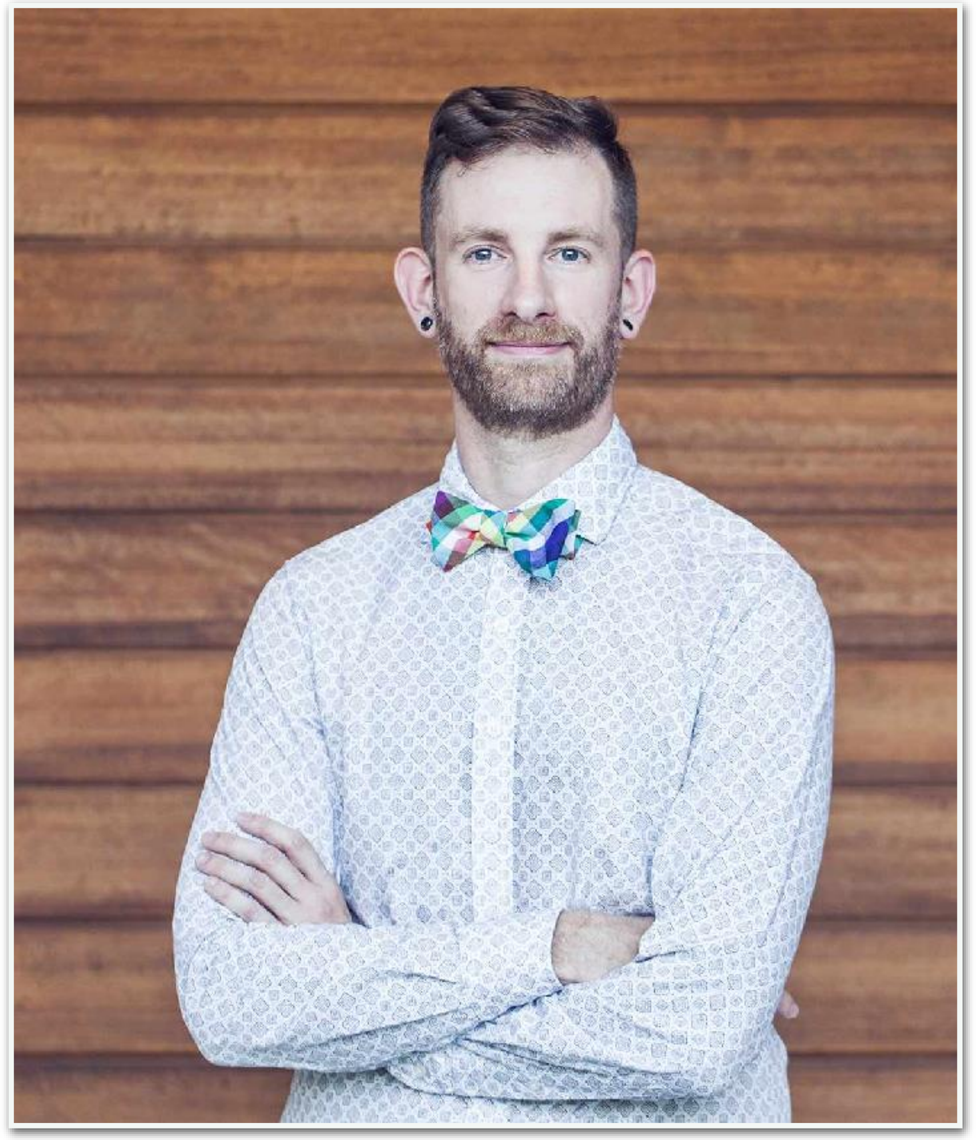
https://medium.com/analytics-vidhya/split-apply-combine-strategy-for-data-mining-4fd6e2a0cc99

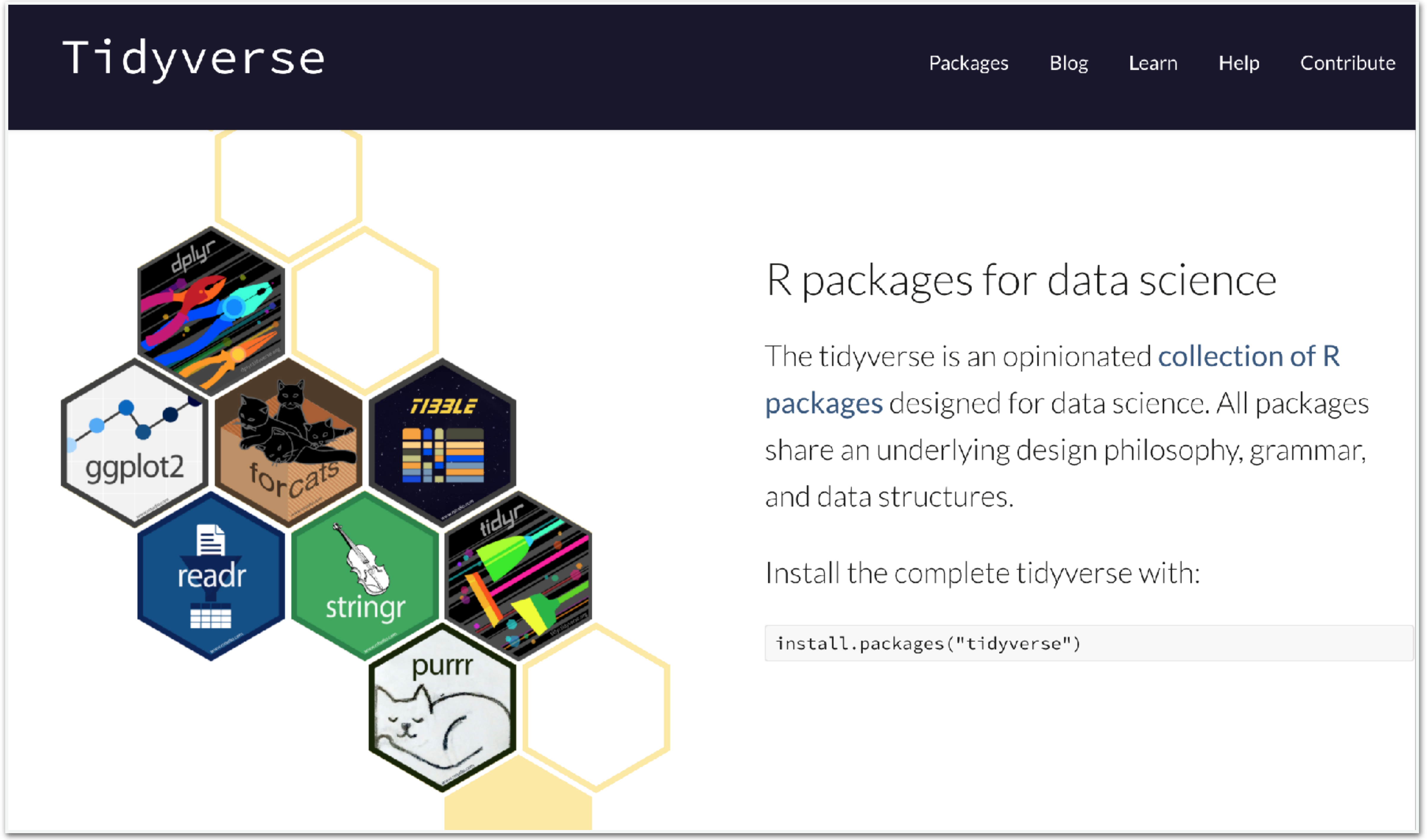
Task 1 — Build the analysis table

The question on the whiteboard

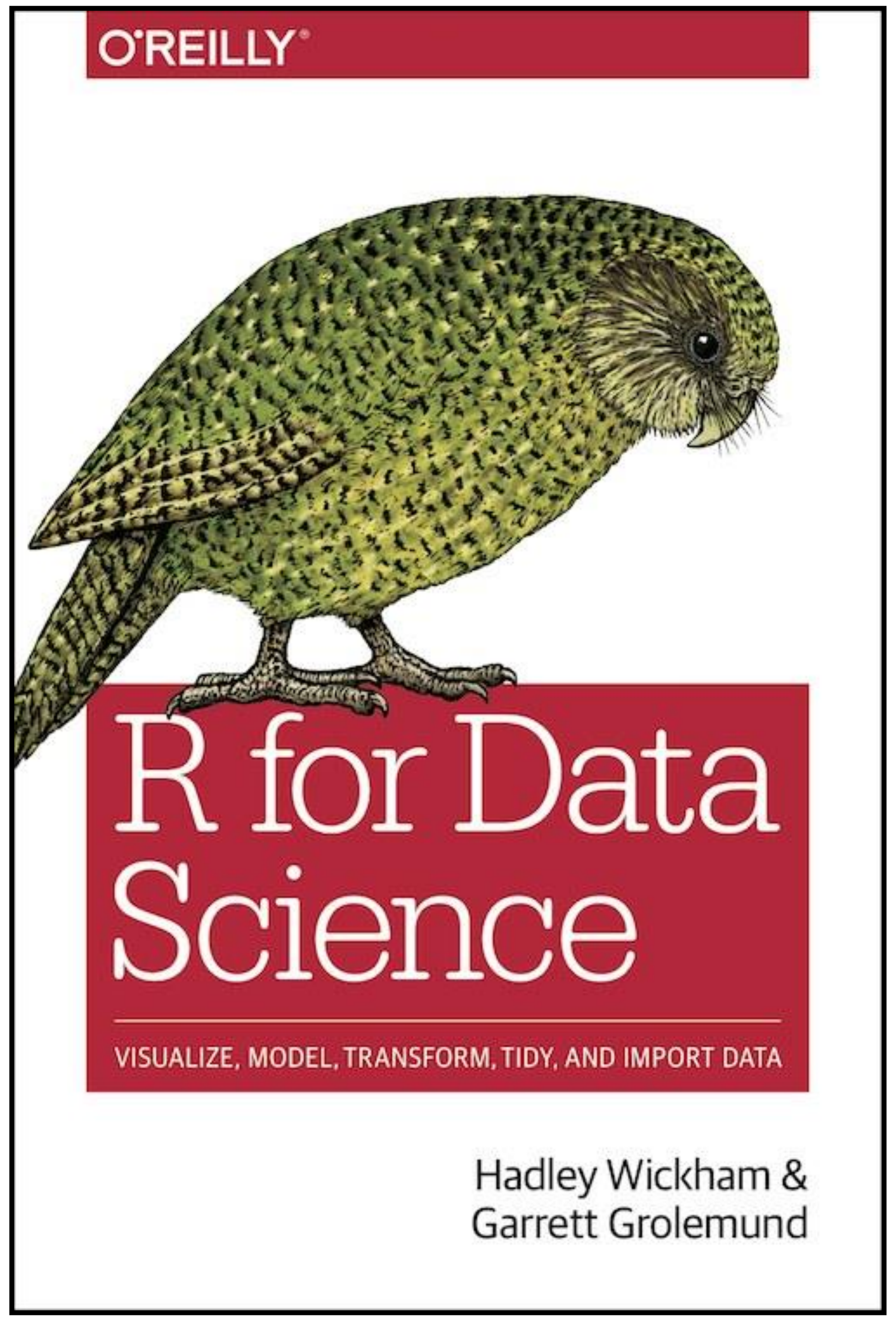
The question	Which of these 32 countries had the worst winter of 2021-22?
The result	One table, with one row per country, and one rate you chose.
Save it	The afternoon starts from this table, so write it out.
Post it	Your table and one-sentence comparison on the wrangle Padlet. Then compare with your neighbour.

The tidyverse



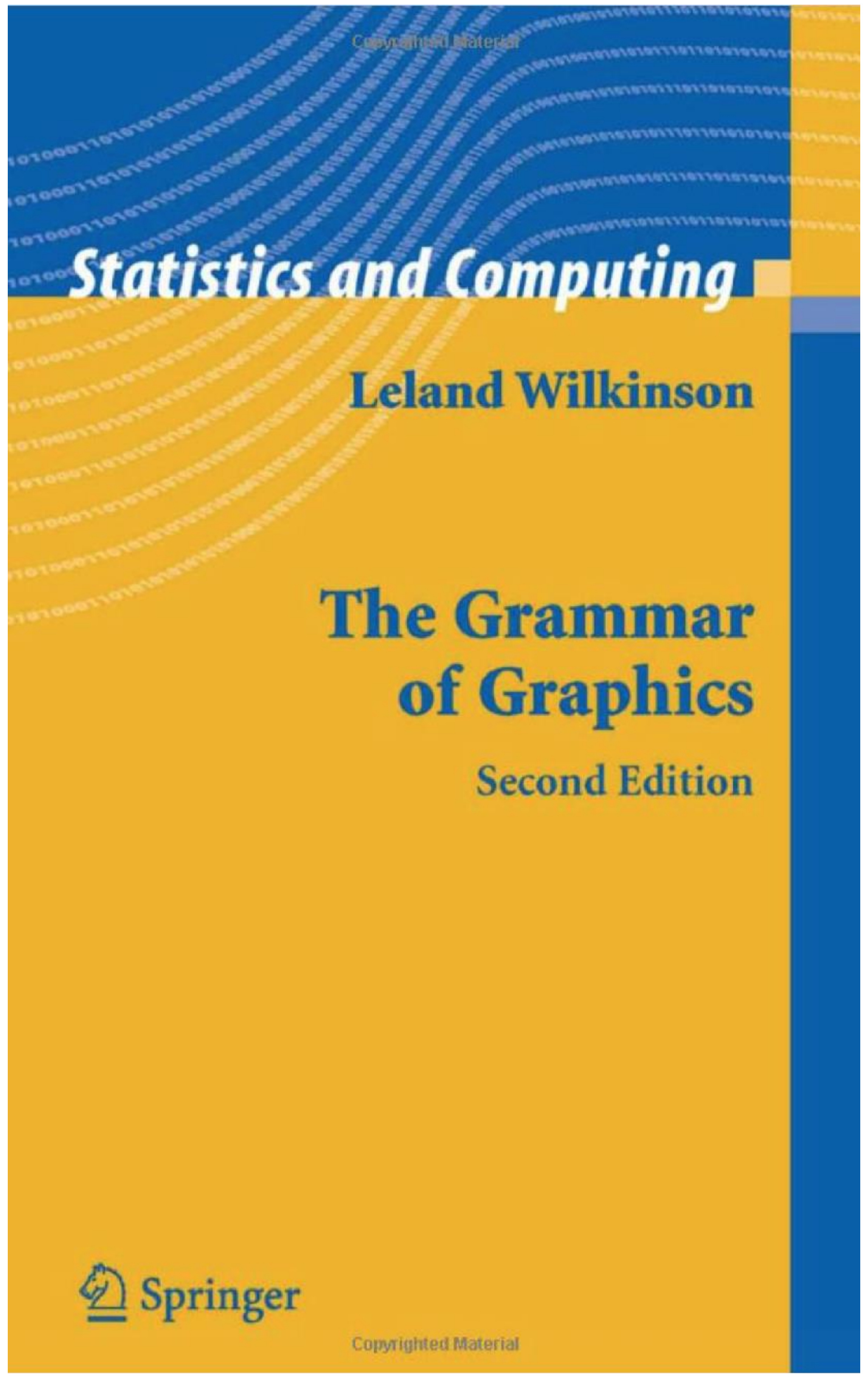


www.tidyverse.org



r4ds.had.co.nz

A layered grammar of graphics

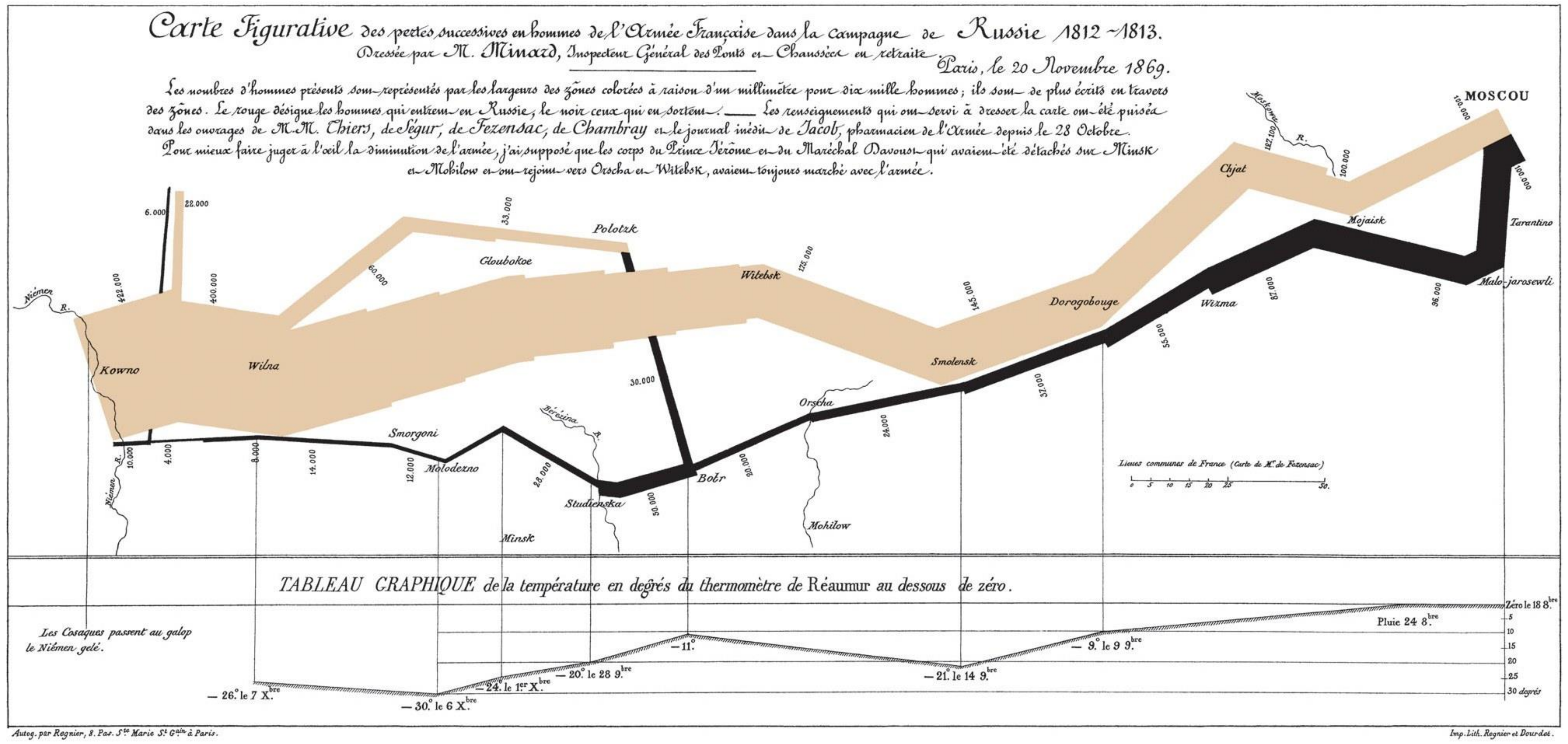


A description of the fundamental features that underlie all statistical graphics

A toolkit implementing the grammar of graphics in R

A layered grammar of graphics

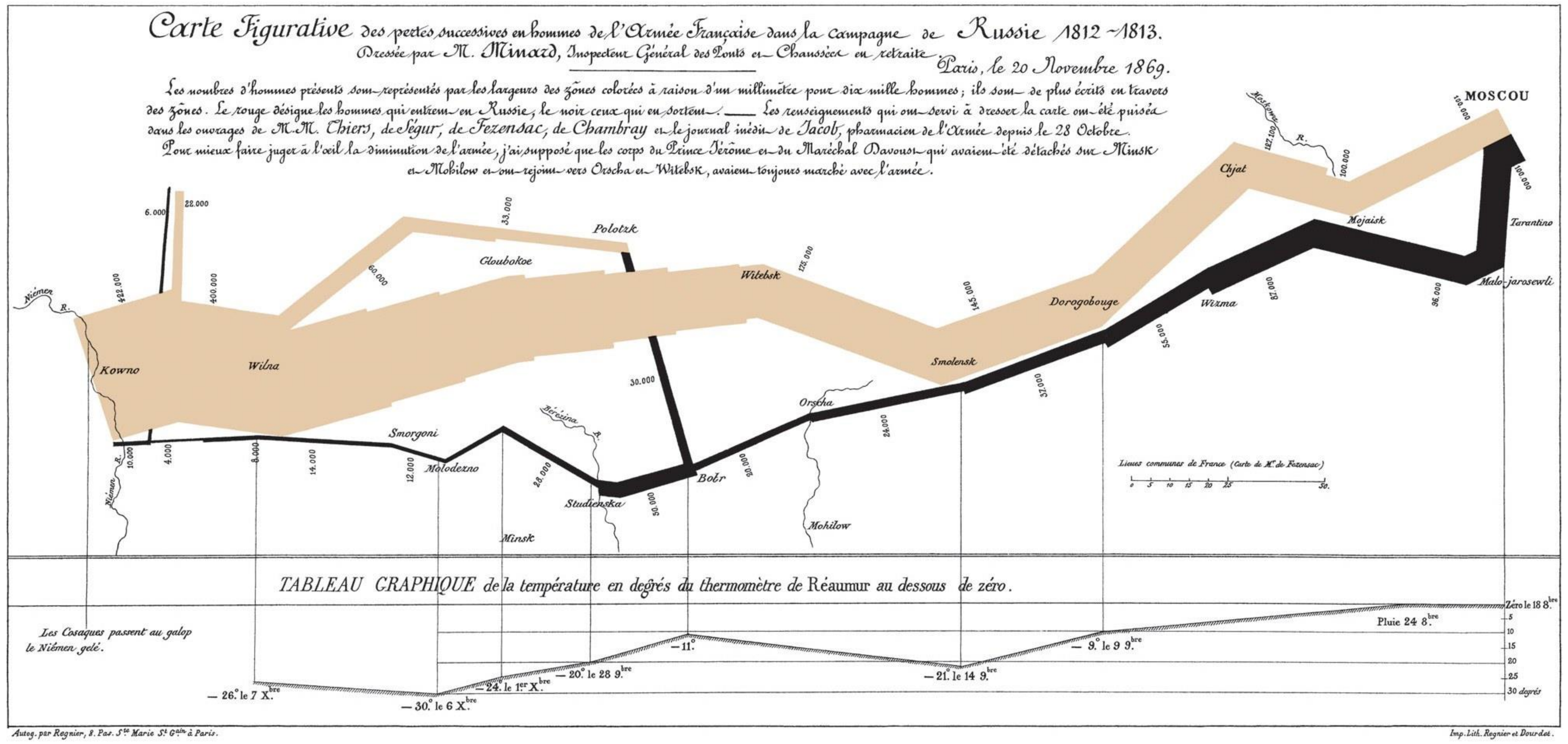
Original



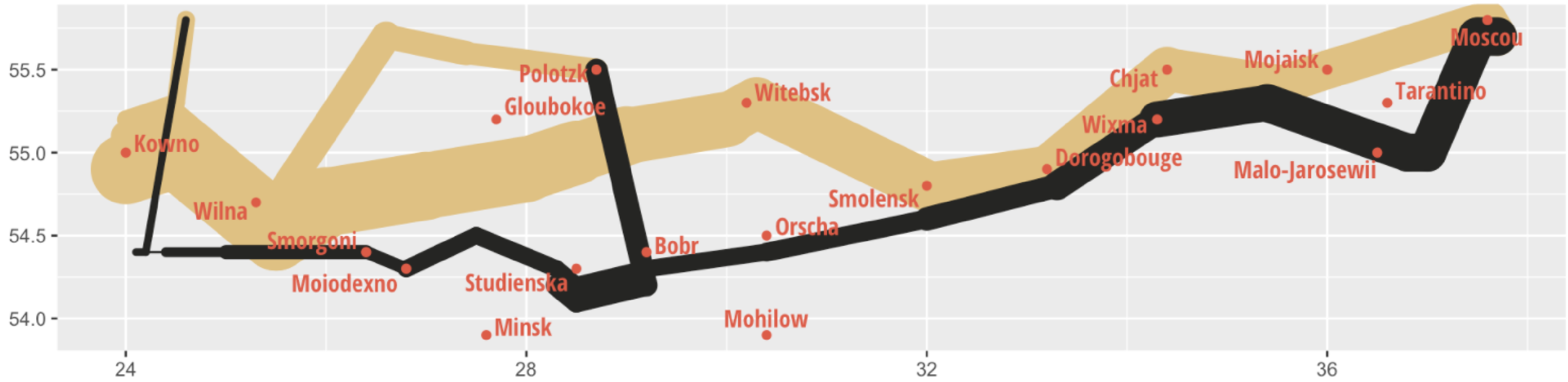
https://www.andrewheiss.com/blog/2017/08/10/exploring-minards-1812-plot-with-ggp

A layered grammar of graphics

Original



ggplot2 version



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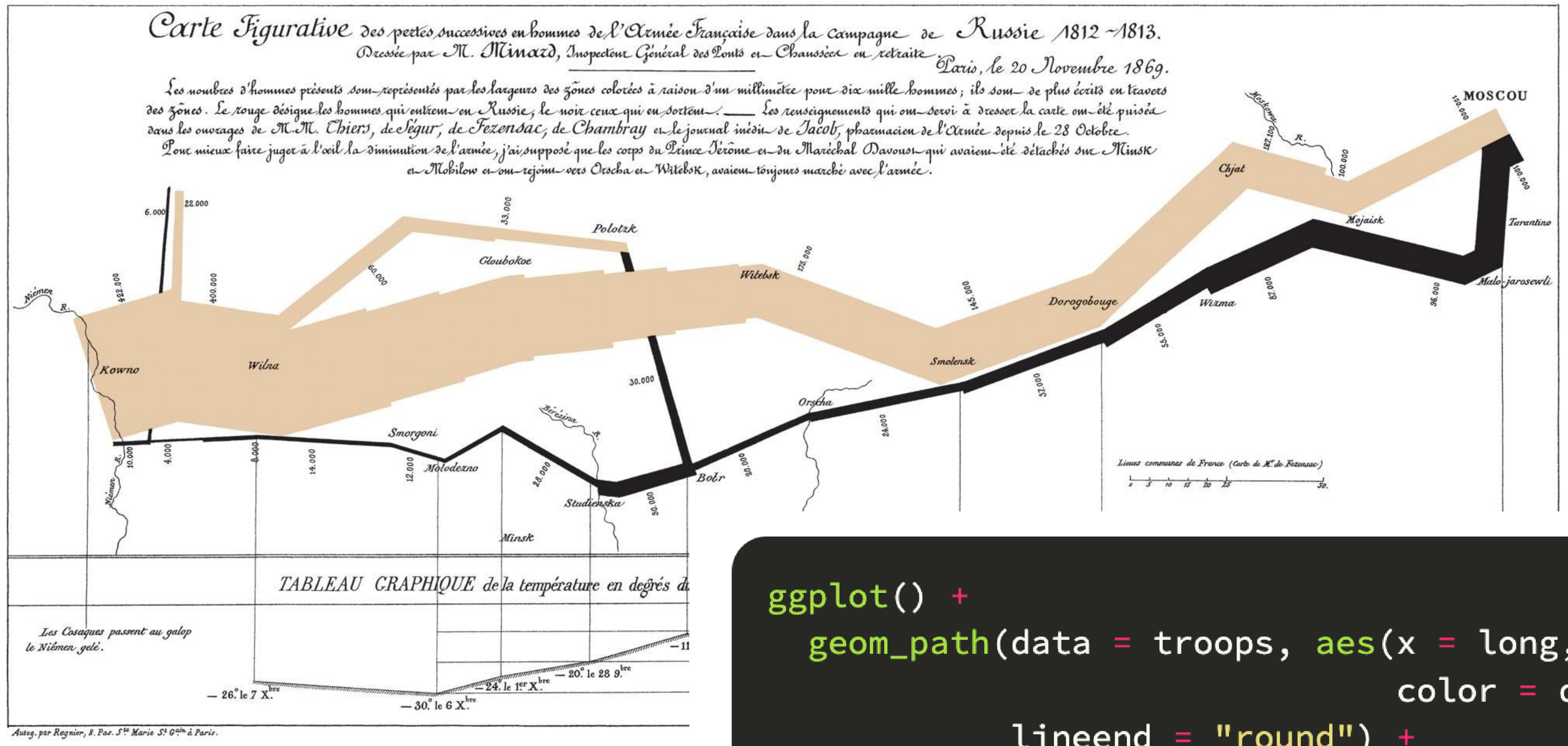
A layered grammar of graphics

Original



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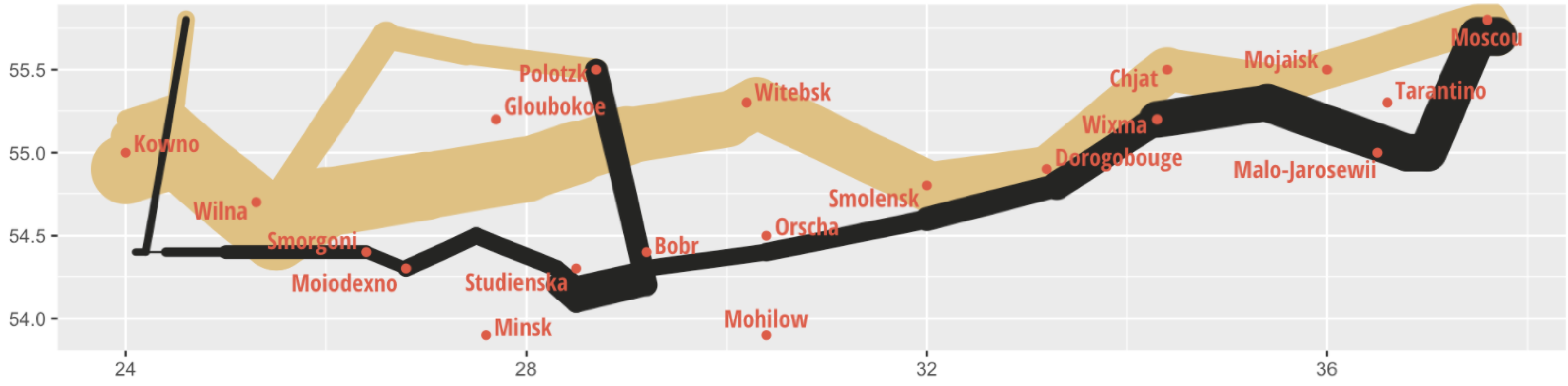



```r
ggplot() +
  geom_path(data = troops, aes(x = long, y = lat, group = group,
                               color = direction, size = survivors),
            lineend = "round") +
  geom_point(data = cities, aes(x = long, y = lat),
             color = "#DC5B44") +
  geom_text_repel(data = cities, aes(x = long, y = lat, label = city),
                  color = "#DC5B44", family = "Open Sans Condensed Bold") +
  scale_size(range = c(0.5, 15)) +
  scale_colour_manual(values = c("#DFC17E", "#252523")) +
  labs(x = NULL, y = NULL) +
  guides(color = FALSE, size = FALSE)
```

https://www.andrewheiss.com/blog/2017/08/10/exploring-minards-1812-plot-with-ggp

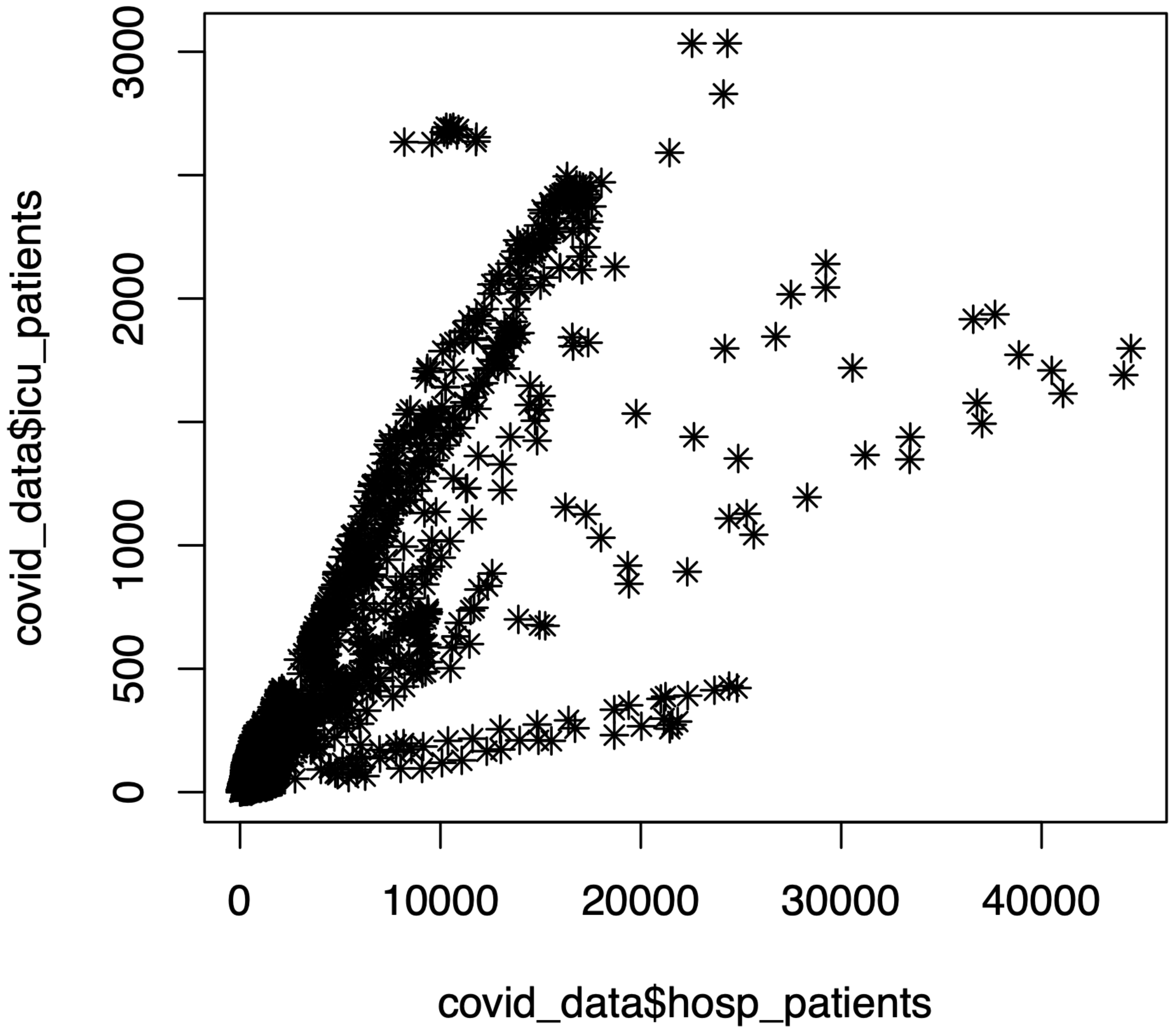
A layered grammar of graphics

data MAPPED TO **aes**thetic properties OF **geom**etric objects

ggplot2

https://rfortherestofus.com/2019/07/a-short-overview-of-the-grammar-of-grap

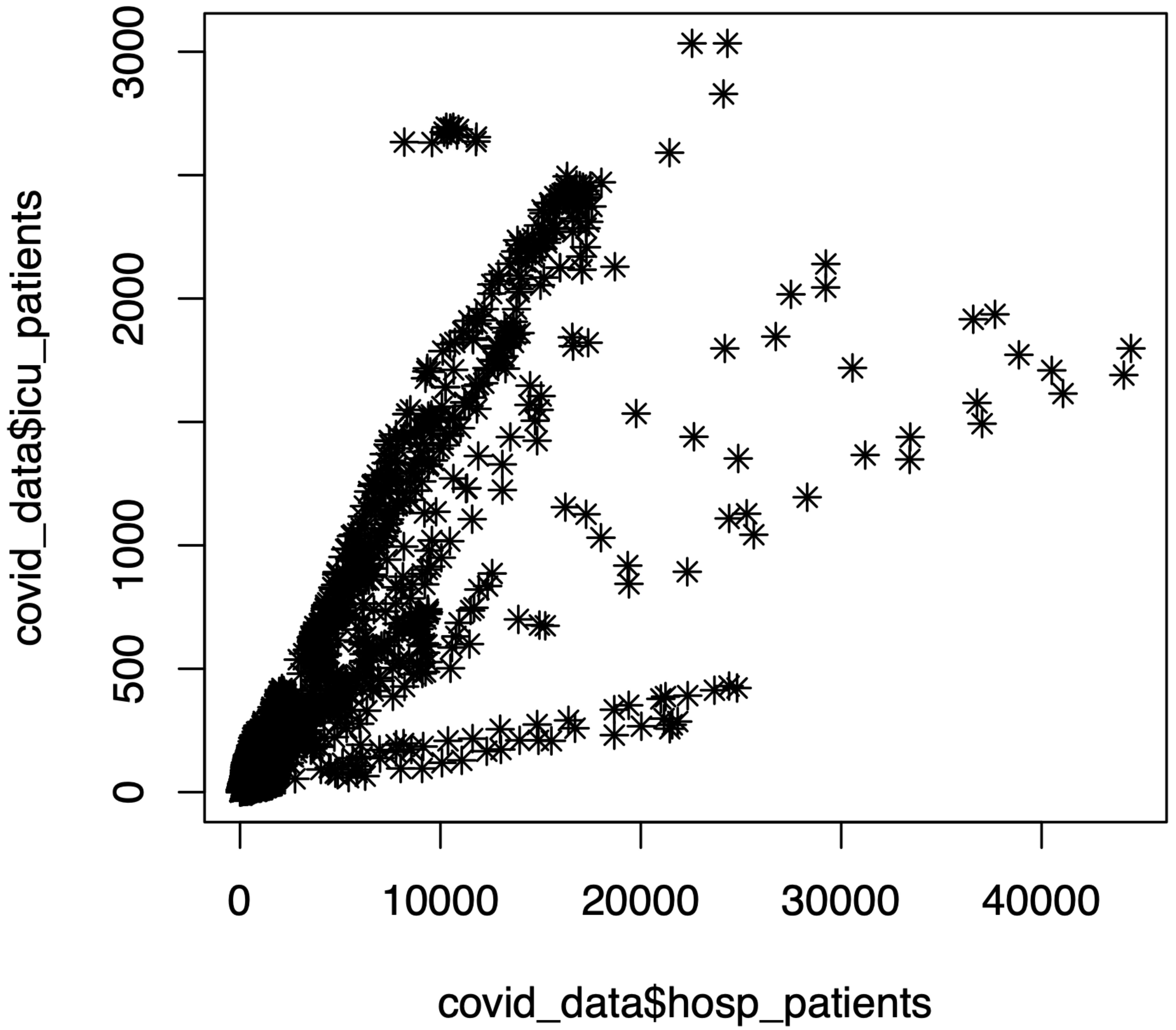
Anatomy of a scatter plot



data MAPPED TO aesthetic properties OF geometric objects

```
plot(x = covid_data$hosp_patients,
     y = covid_data$icu_patients,
     pch = 8,
     type = “p”)
```

Anatomy of a scatter plot



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Data

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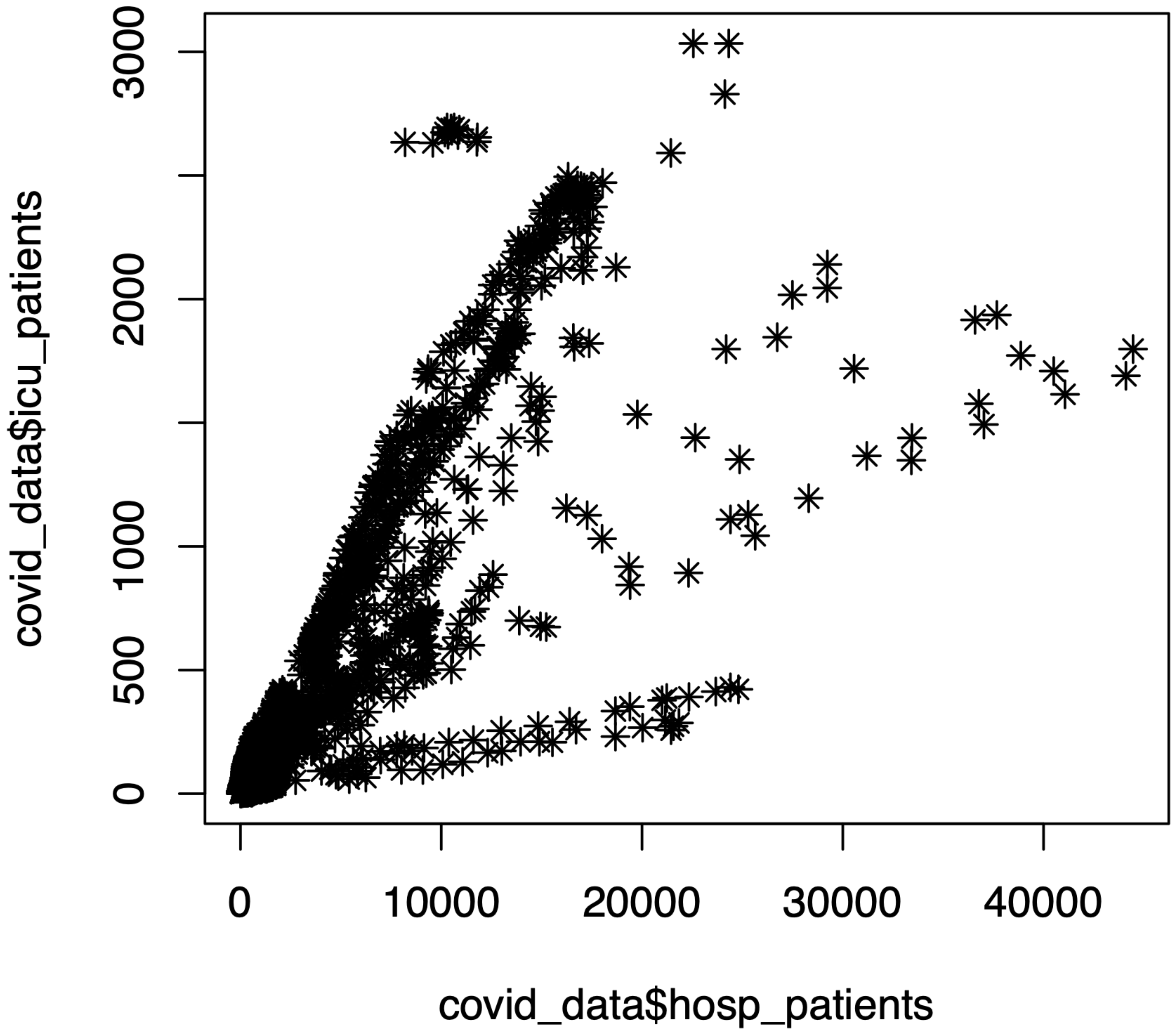
Anatomy of a scatter plot



data MAPPED TO **aes**thetic properties OF **geom**etric objects

Data

Aesthetic properties

```
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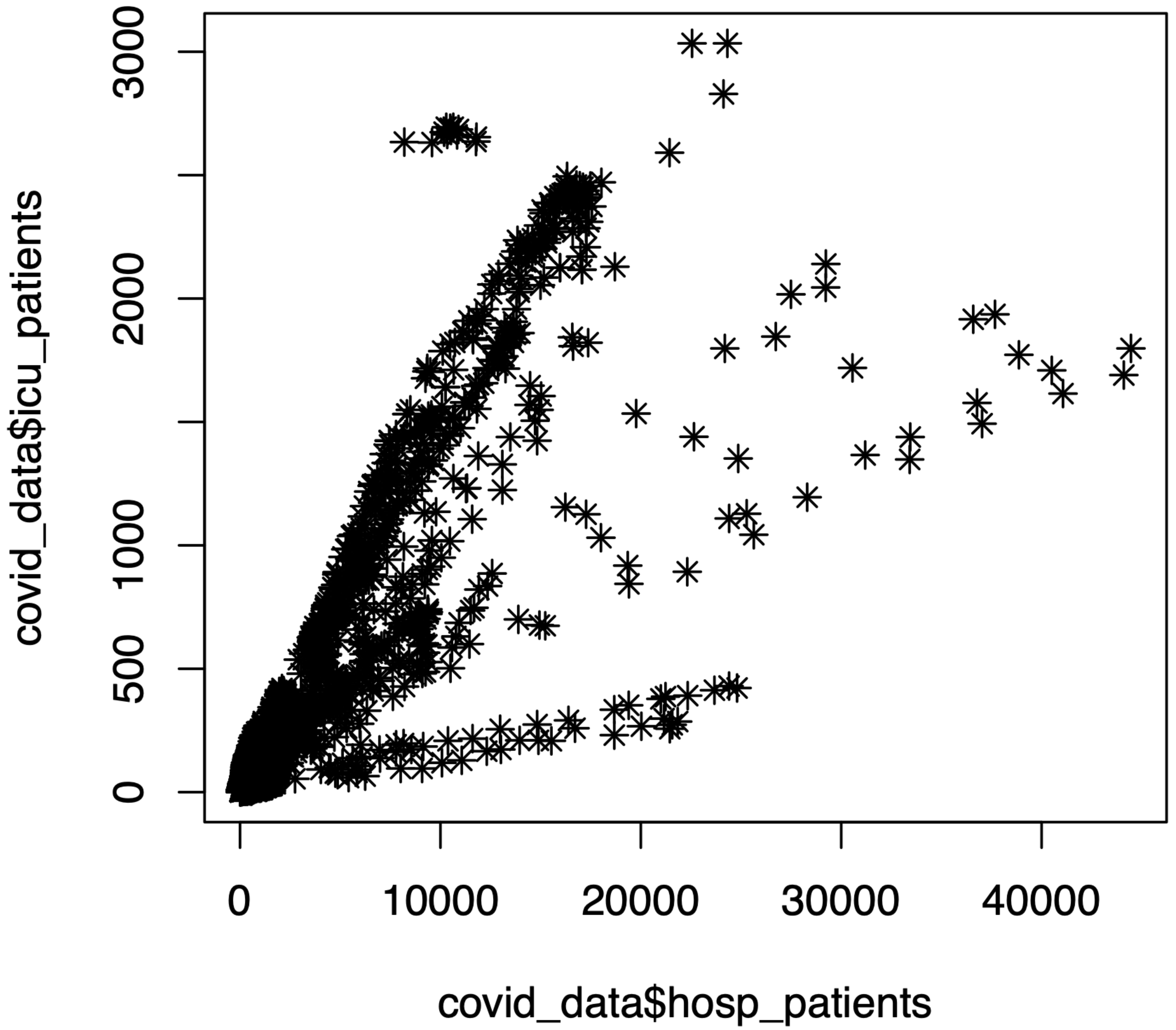

data MAPPED TO aesthetic properties OF geometric objects

Data

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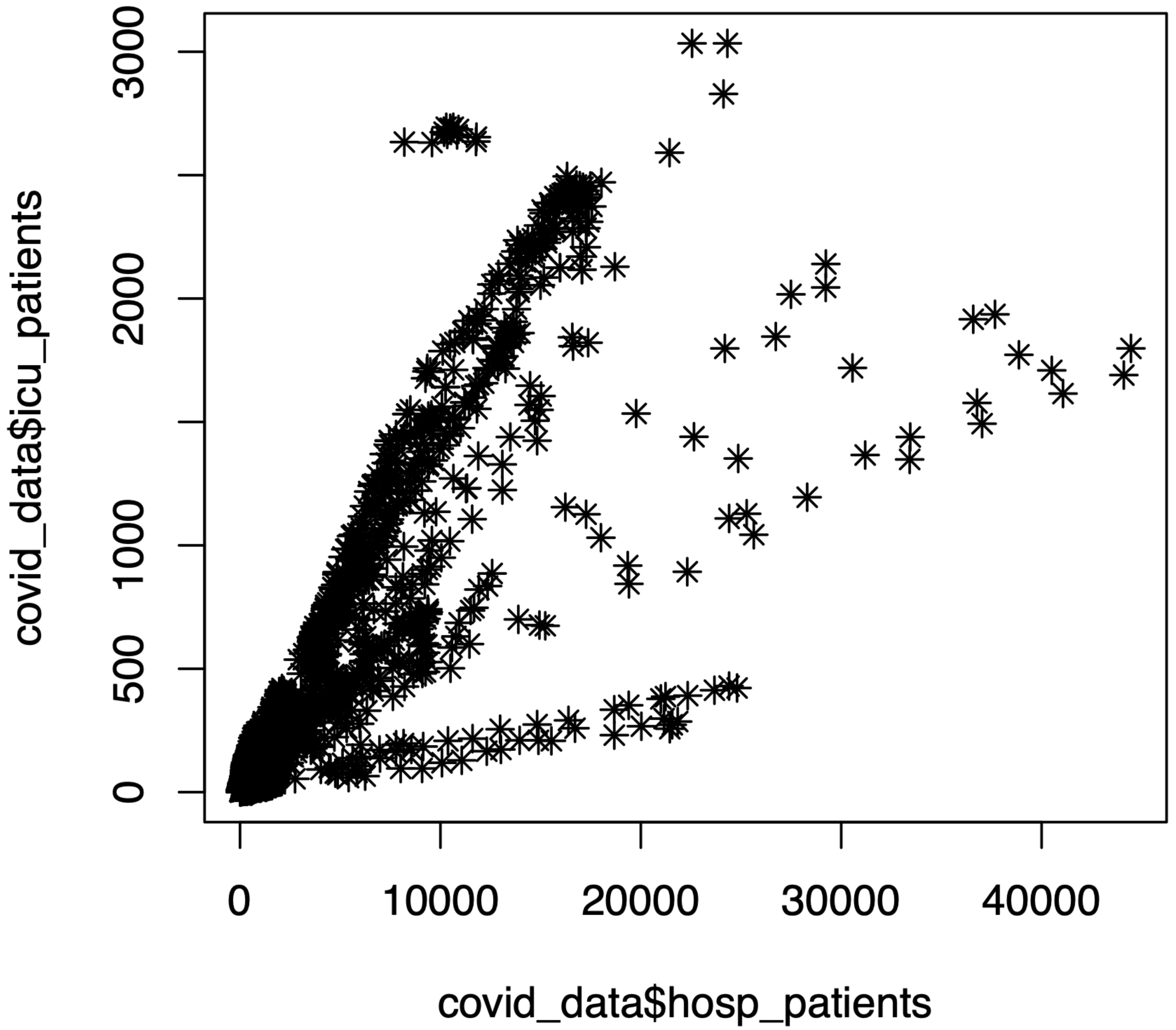
```

Aesthetic properties

Variables

Anatomy of a scatter plot



data MAPPED TO aesthetic properties OF geometric objects

Data

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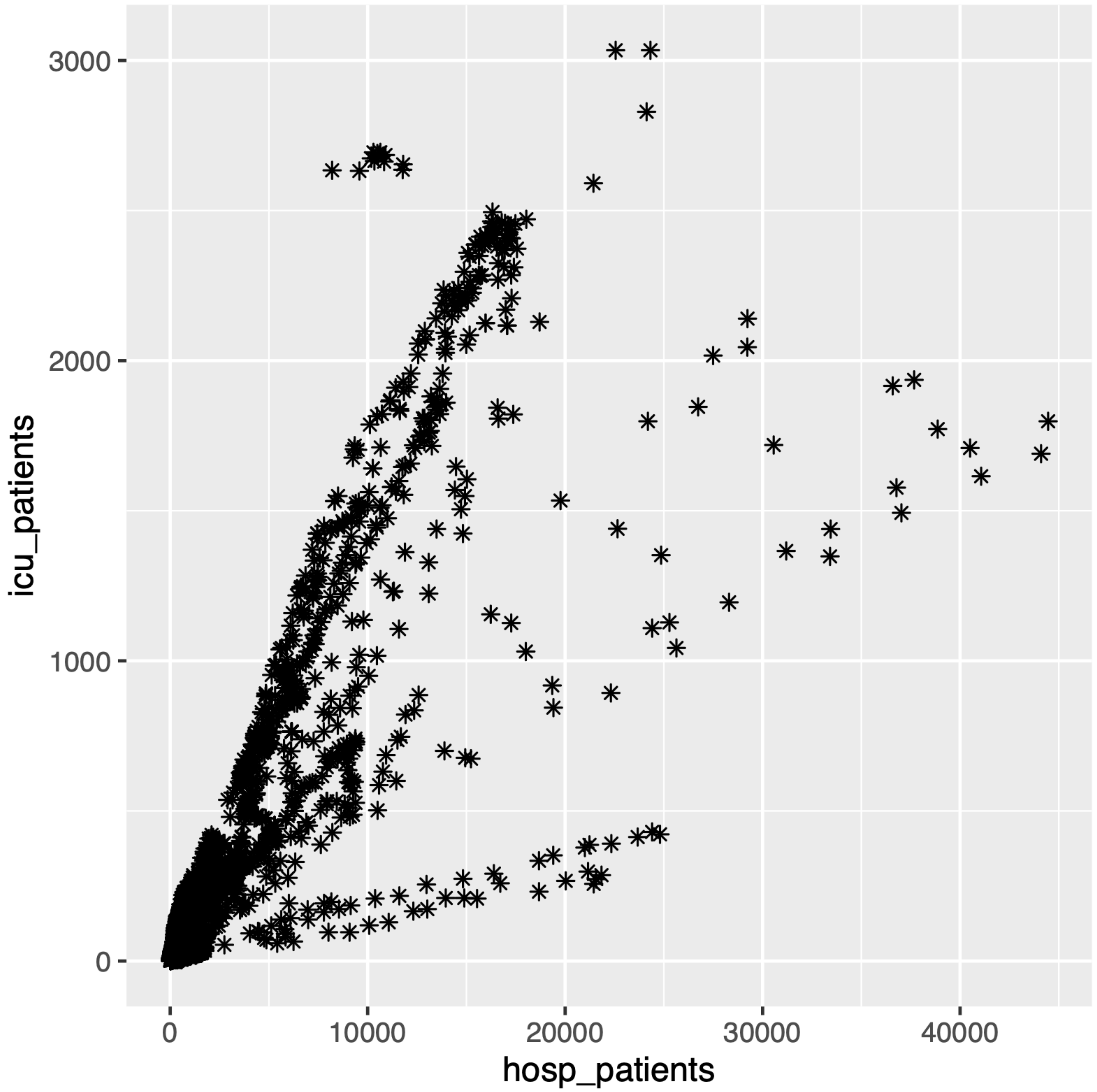
plot(x = covid_data$hosp_patients,
     y = covid_data$icu_patients,
     pch = 8,
     type = “p”)
```

Aesthetic properties

Variables

Geomet
ric
object

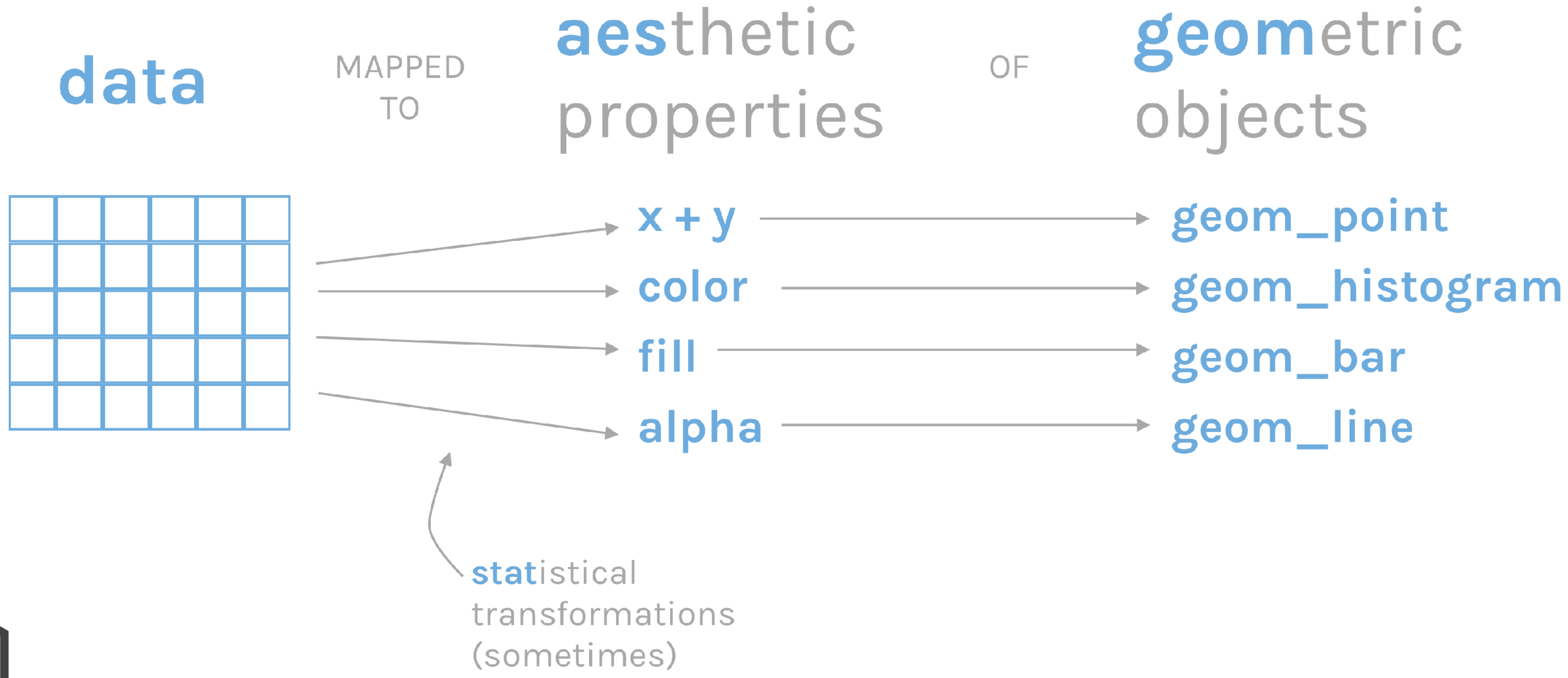
Anatomy of a simple scatter plot



data MAPPED TO **aes**thetic properties OF **geom**etric objects

```
ggplot(
  data = covid_data,
  mapping = aes(
      x = hosp_patients,
      y = icu_patients)
  ) +
geom_point(shape = 8)
```

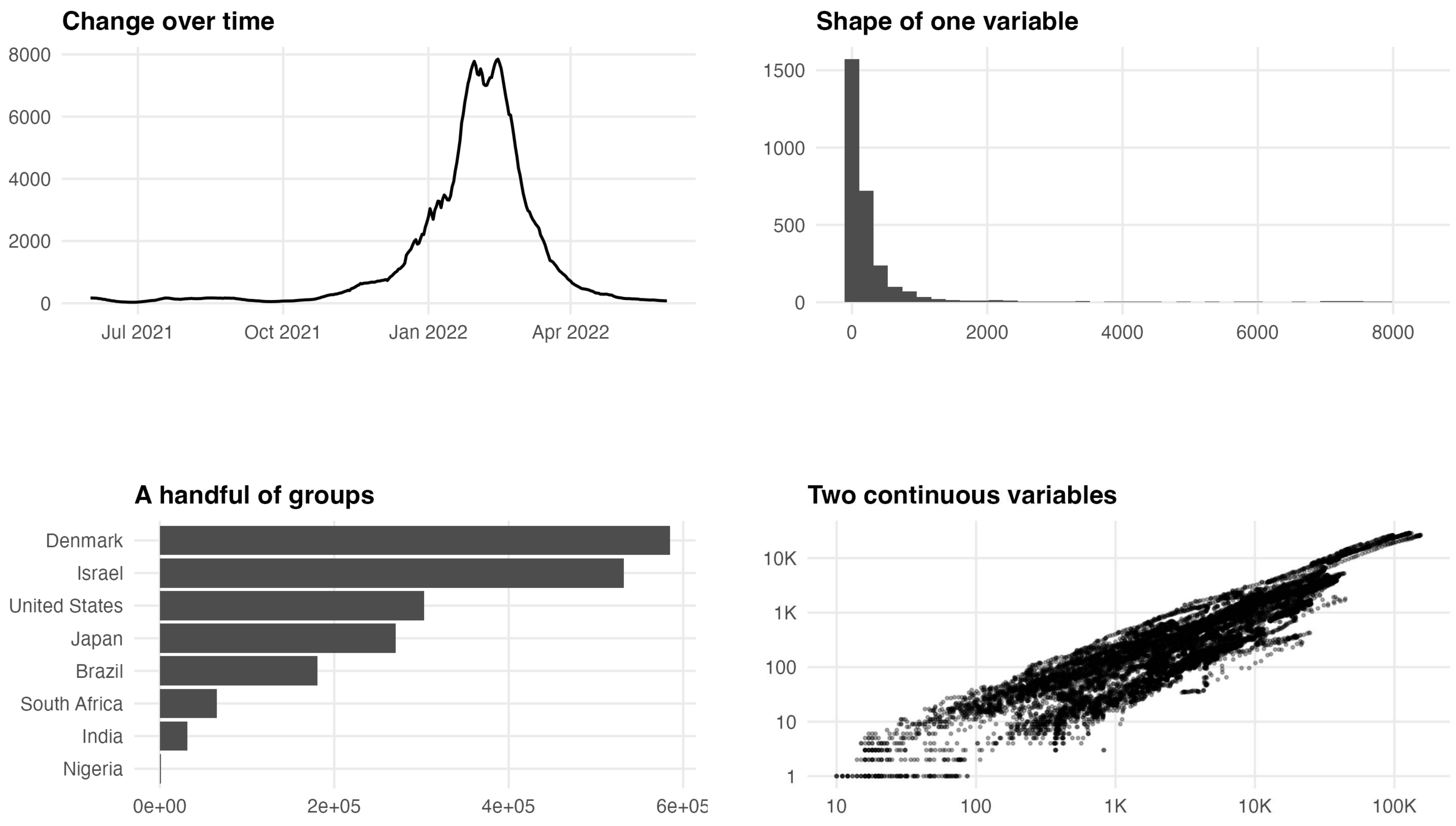
A layered grammar of graphics



https://rfortherestofus.com/2019/07/a-short-overview-of-the-grammar-of-grap

How do I choose a chart type?

Choose the comparison first. The geometry follows.

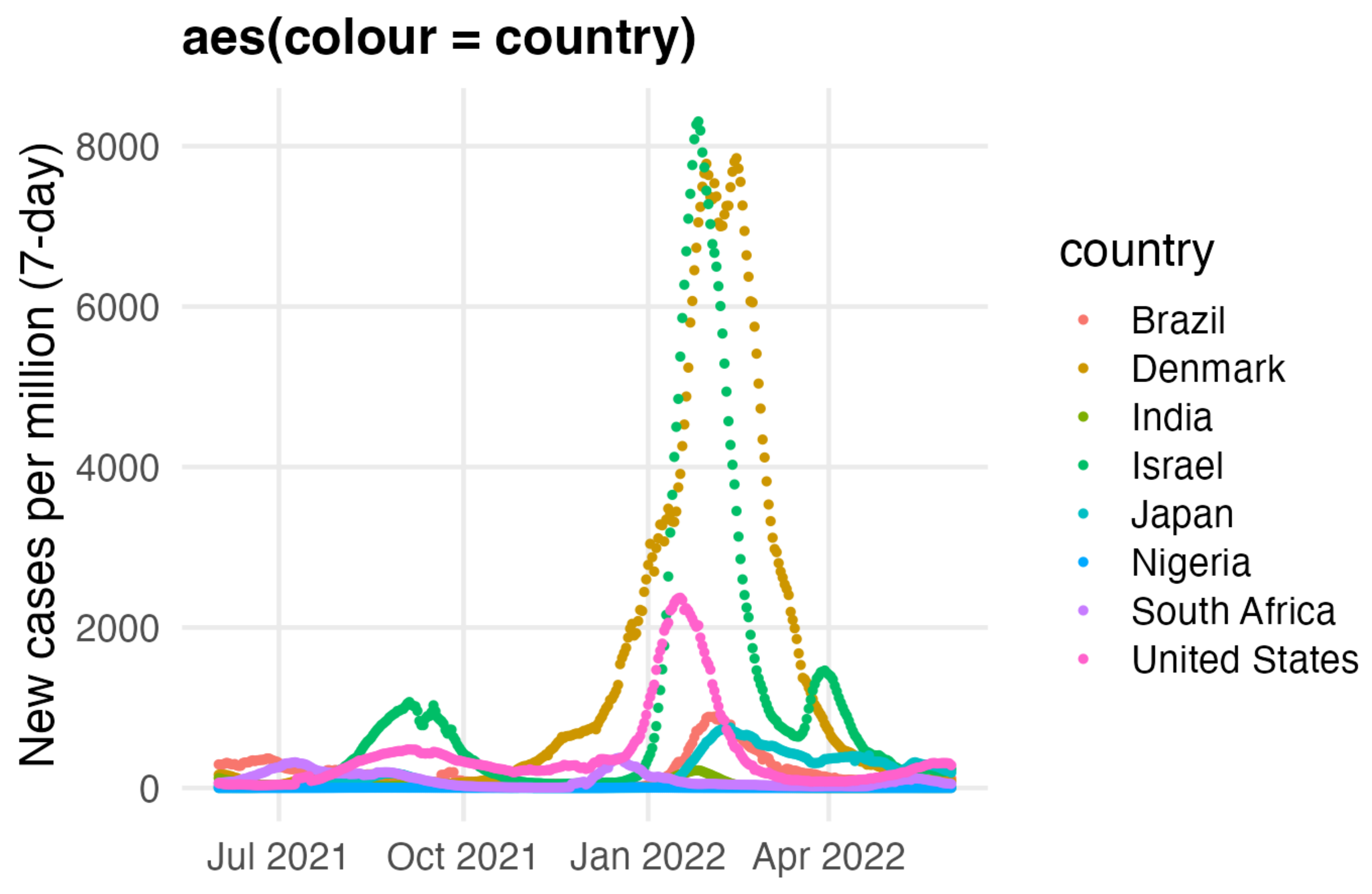


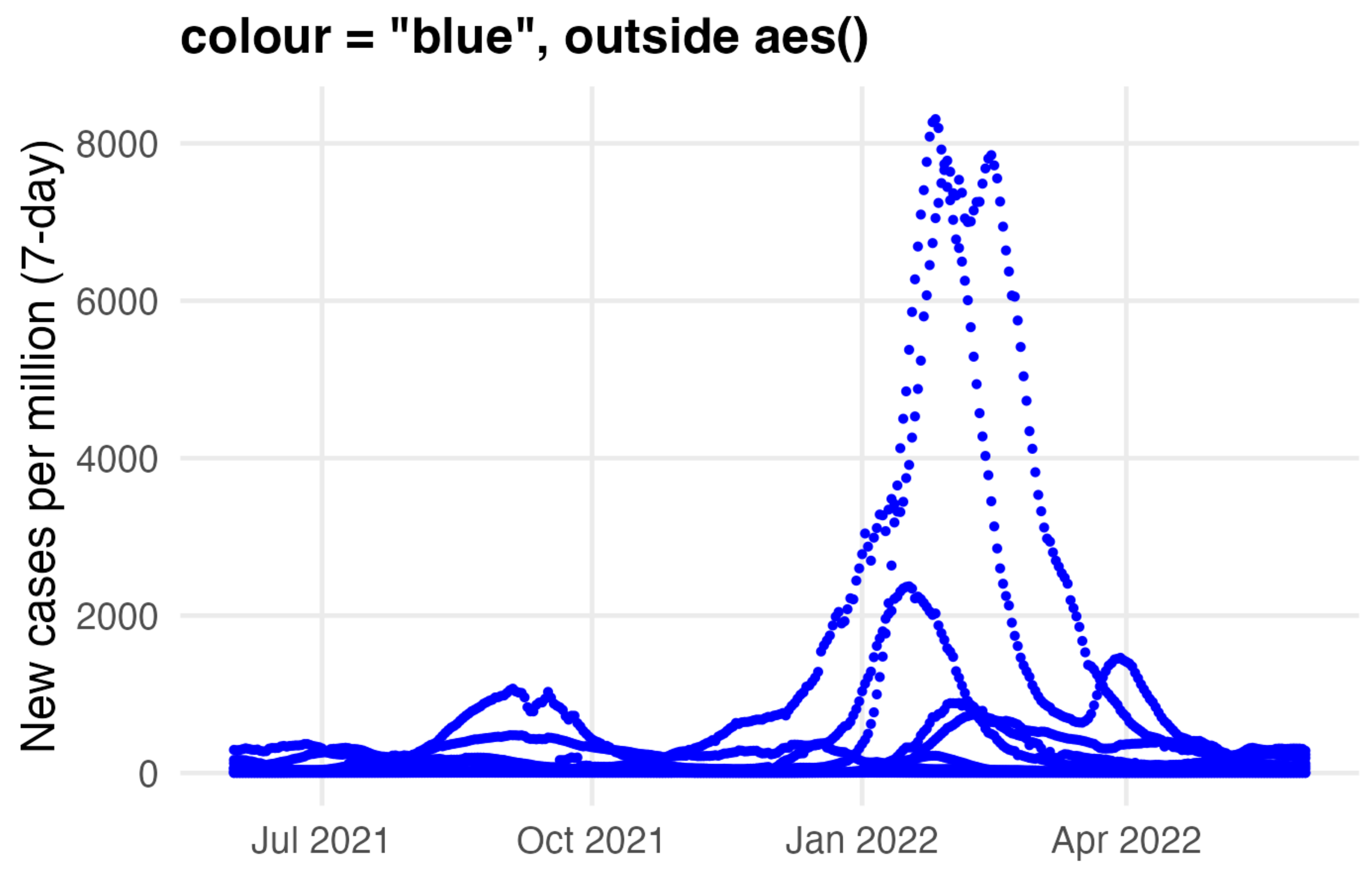
change over time	line
the shape of one variable	histogram
a handful of groups	bar or point
two continuous variables	scatter

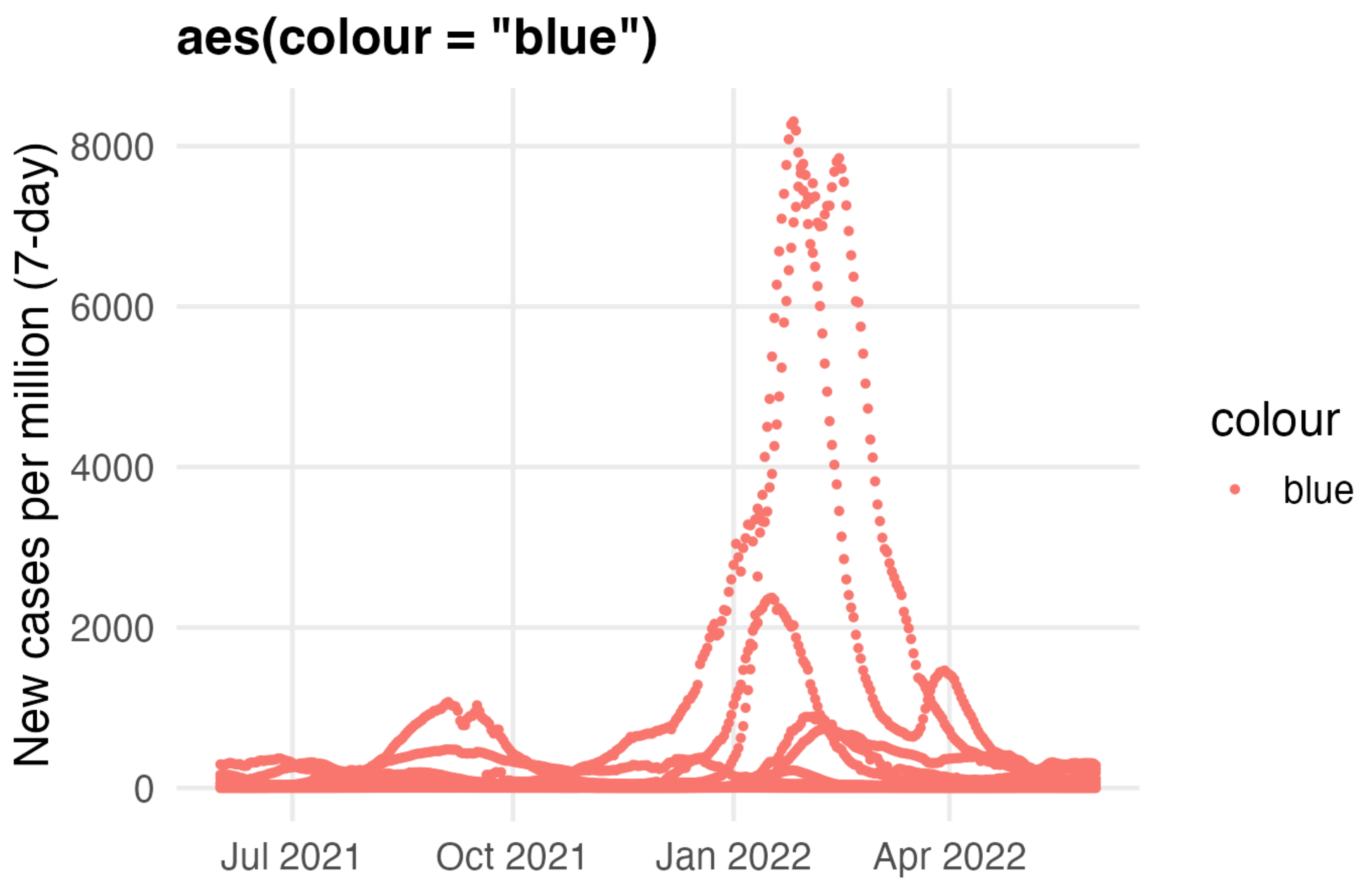
Which of the countries in this dataset had the worst winter of 2021-22, and by what measure?

`https://padlet.com/ucph/fundamentals_2026_module2_dataviz`

Inside aes() or outside?







aes(colour = country)

Colour carries data. Legend appears.

colour = "blue"

A constant. All points blue, no legend.

aes(colour = "blue")

One group called “blue”. Everything salmon-red.

The third is not an error. It is ggplot doing exactly what you asked: a one-level variable whose only value is the string “blue”, mapped to colour.

aes() maps a variable to an aesthetic. Everything outside sets a constant.

Task 2

Task 2

One figure from your morning table that answers your stated question. Keep it simple.

Then save it and post it to the dataviz Padlet:

```
ggsave("figures/<yourname>.png", width = 7, height = 5, dpi = 150)
```

Review the figure

1. Does it answer a stated question?
2. Is the figure easy to read and interpret?
3. Are units, denominators and axes labelled accordingly?
4. What does it hide?
5. What would you change to make it clearer?

One comment per figure, on your neighbour's post